



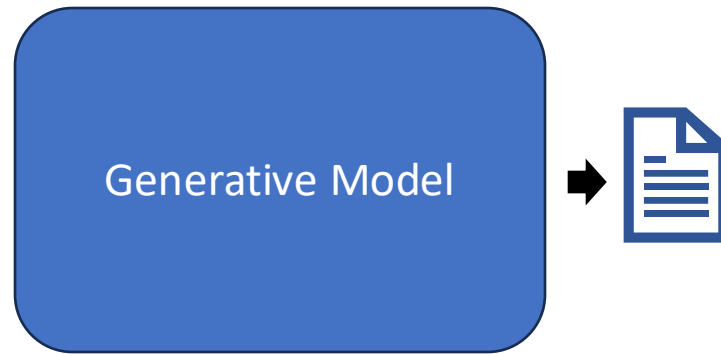
Lecture 7:

Multimodal Language Models

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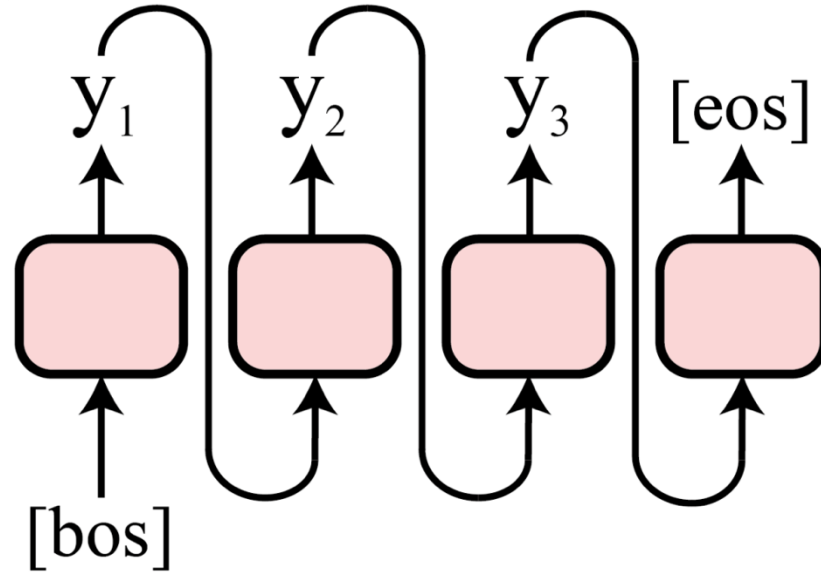
Generative models

- Generative models aim to model data probability $p(y)$



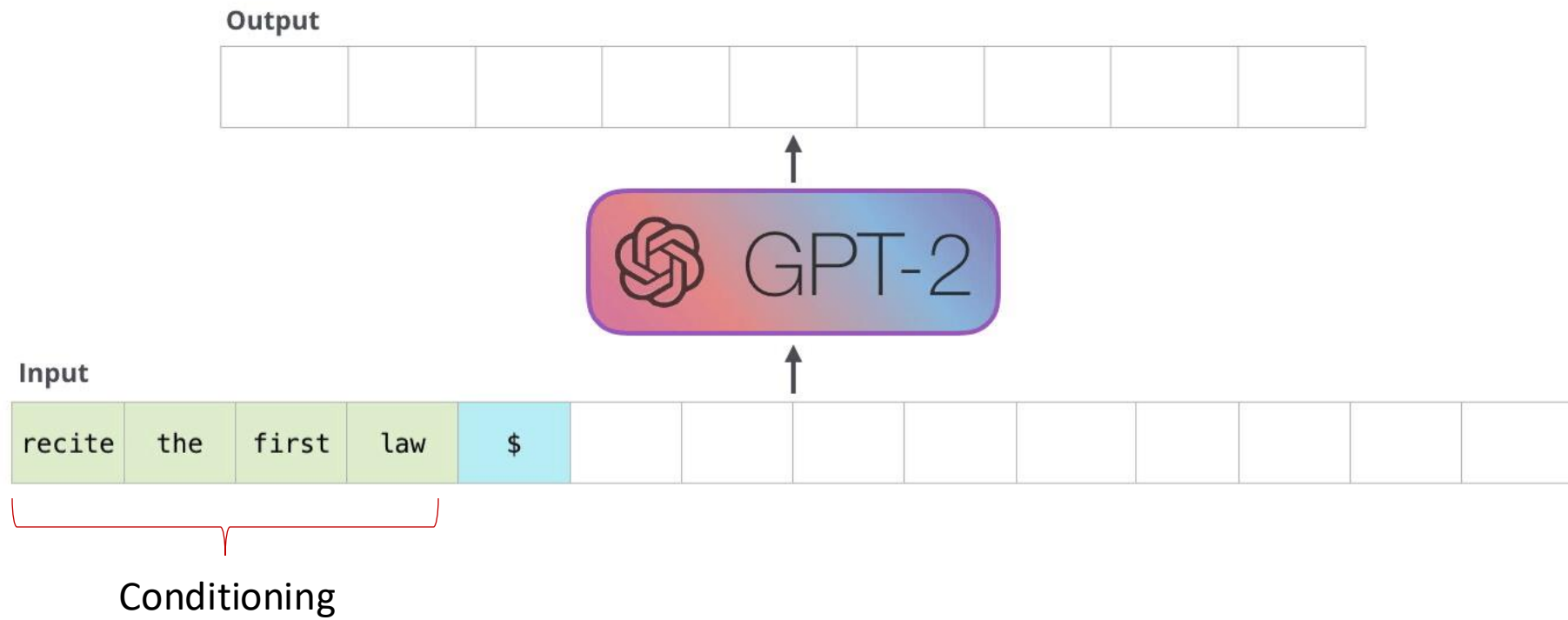
Auto-regressive modeling

$$p(y) = \prod_{i=1}^n p(y_i | y_{j < i})$$

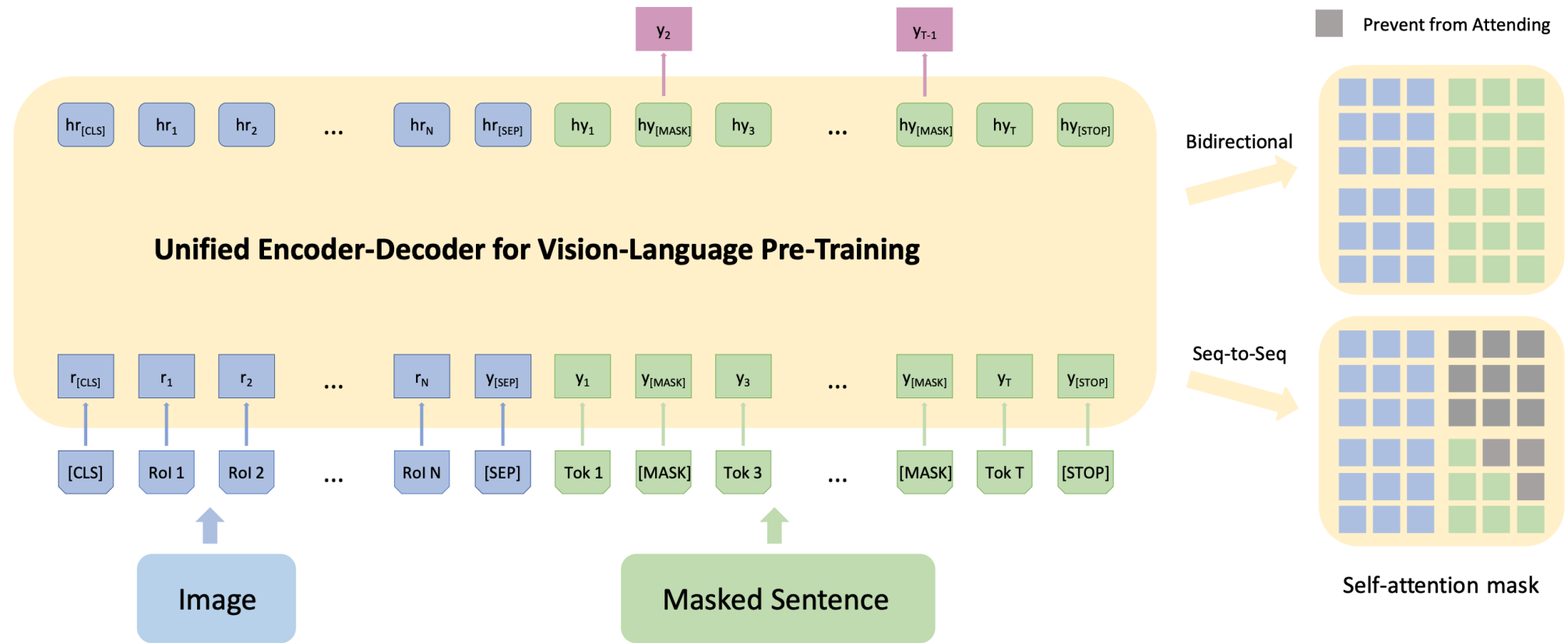


But how to model $p(x|c)$?

Recall: GPT-2

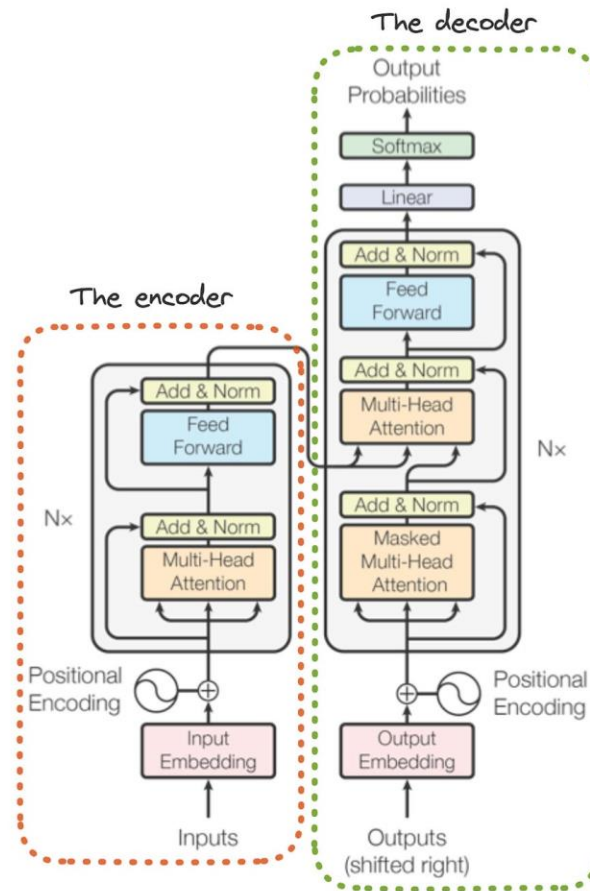


VLP: Adding seq-to-seq masking



The original transformer paper

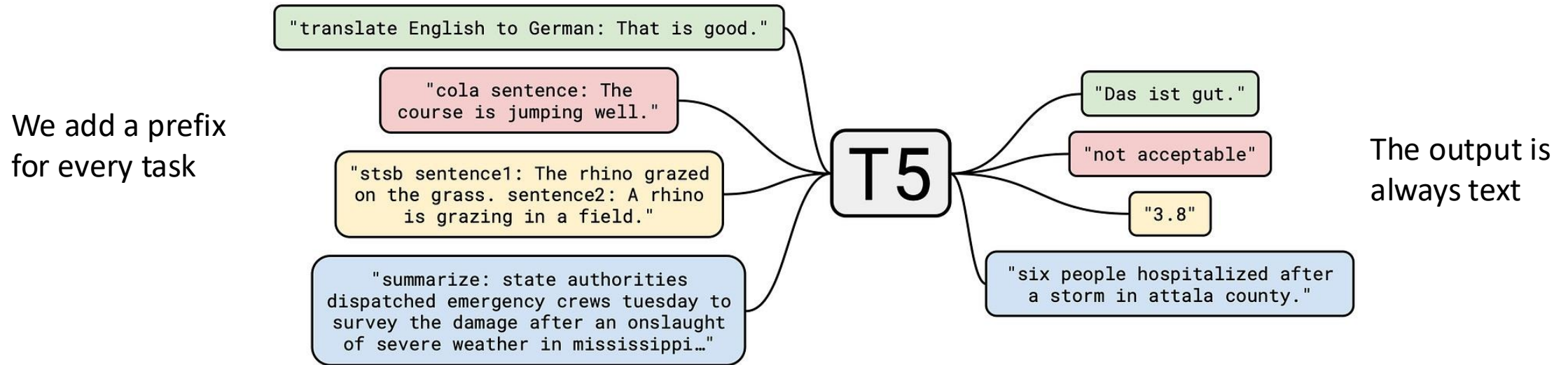
The encoder is great for understanding the prompt conditioning



The decoder is great for language generation

Encoder-decoder architecture

Text-to-Text Transfer Transformer



Pre-training dataset

- T5 also proposed the Colossal Clean Crawled Corpus (C4) dataset
- A relatively clean dataset created using a limited set of filtering rules.
- This dataset was later used as part of the popular MassiveText dataset, which was utilized by models such as Chinchilla and Gopher.

	Disk Size	Documents	Tokens	Sampling proportion
<i>MassiveWeb</i>	1.9 TB	604M	506B	48%
Books	2.1 TB	4M	560B	27%
C4	0.75 TB	361M	182B	10%
News	2.7 TB	1.1B	676B	10%
GitHub	3.1 TB	142M	422B	3%
Wikipedia	0.001 TB	6M	4B	2%

MassiveText

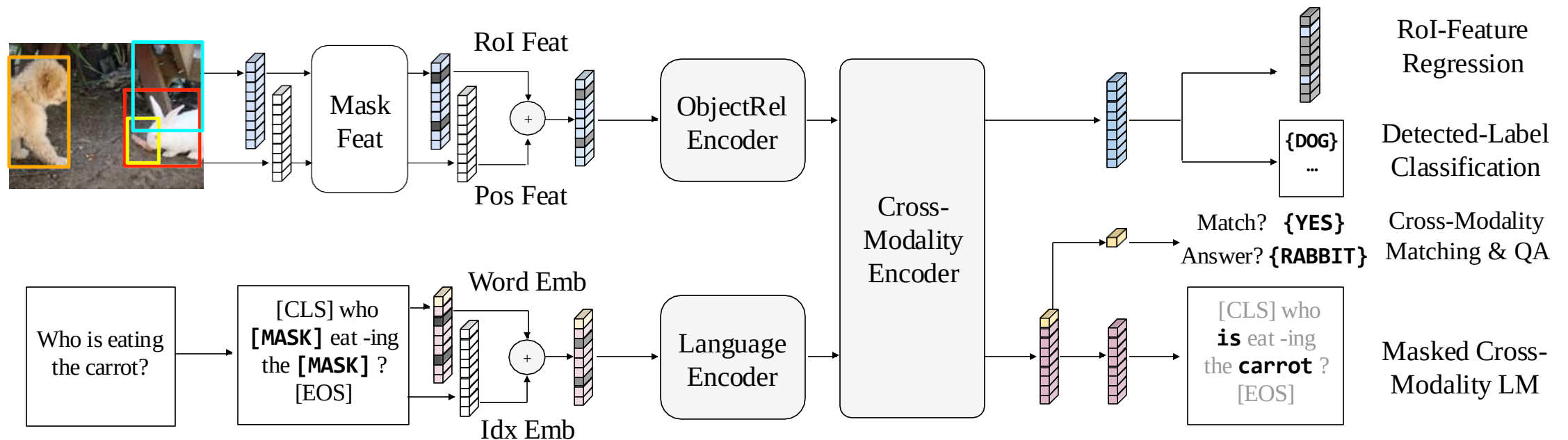
Is text-only possible for classification?

"An issue arises if our model outputs text on a text classification task that does not correspond to any of the possible labels..."

*In this case, we always count the model's output as wrong, though we **never observed this behavior in any of our trained models.**"*

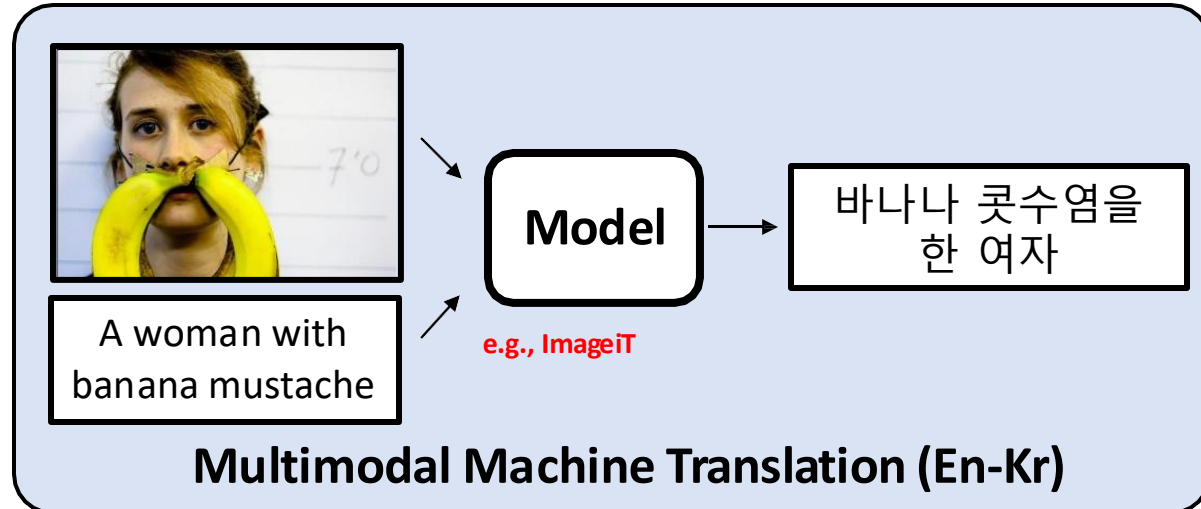
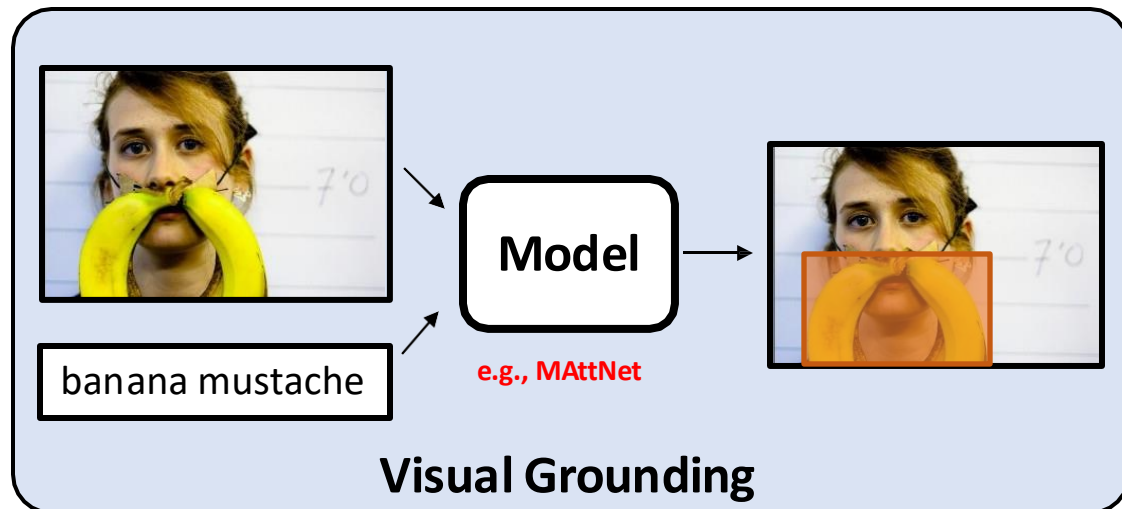
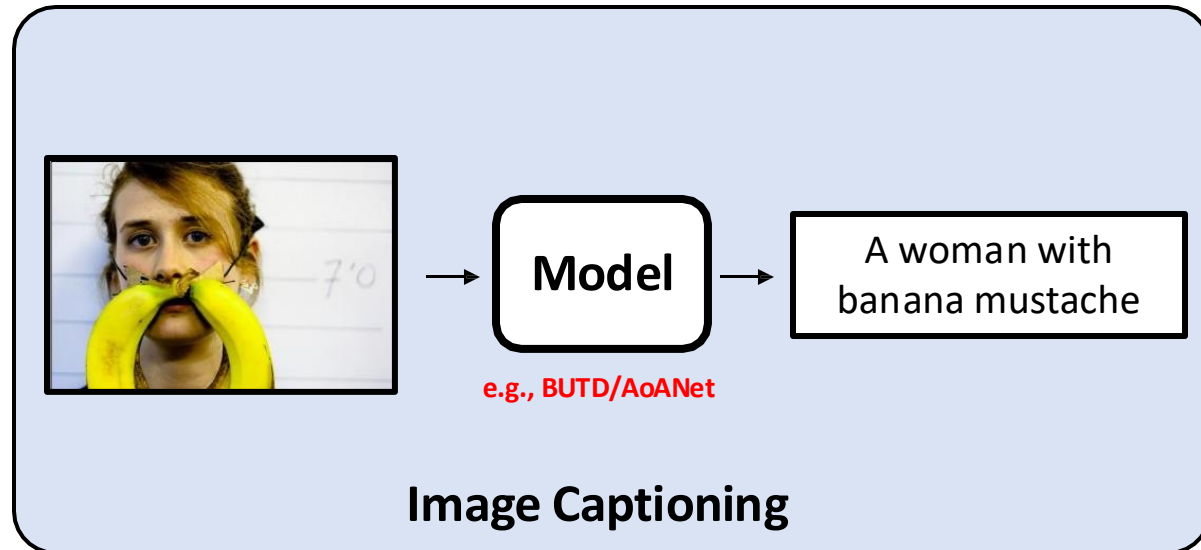
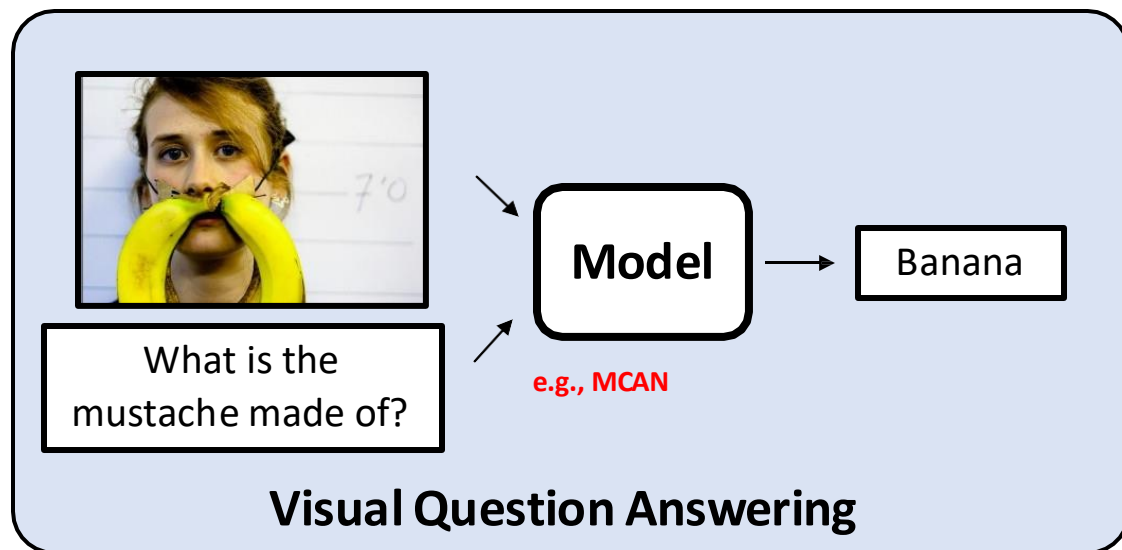
What about multimodal?

We saw LXMERT, a multimodal encoder



- Achieved big gains + sota on several VL tasks such as VQA, GQA, NLVR2, VizWiz, etc.

Diverse Vision-and-Language Tasks



Huang et al., 2019, "Attention on Attention for Image Captioning"

Yu et al., 2018, "MAAttNet: Modular Attention Network for Referring Expression Comprehension"

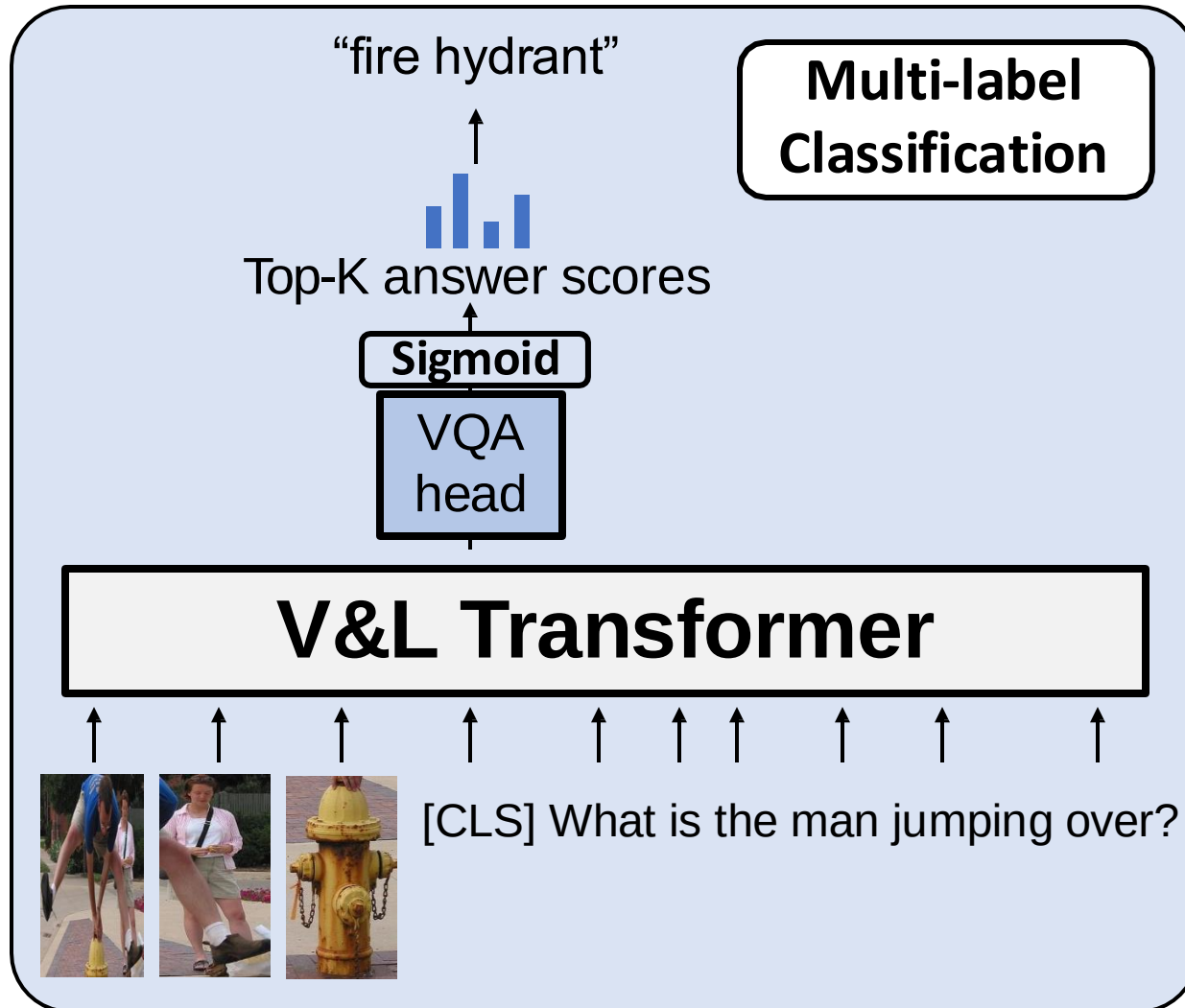
Long et al., 2021, "Generative Imagination Elevates Machine Translation"

Anderson et al., 2018, "Bottom-Up and Top-Down Attention for Image Captioning and Visual Question Answering"

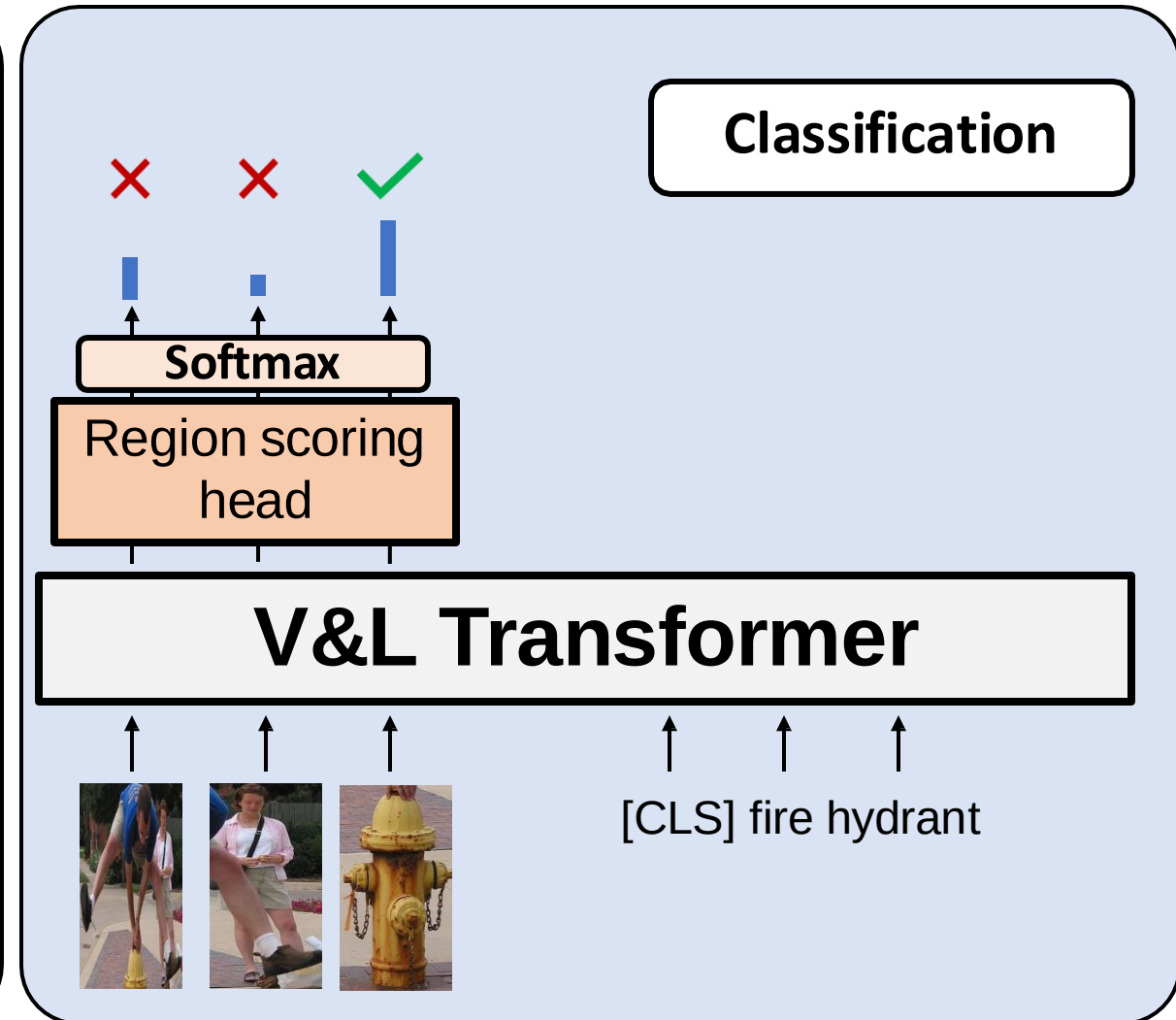
Yu et al., 2019, "Deep Modular Co-Attention Networks for Visual Question Answering"

Task-specific Objectives

Visual Question Answering

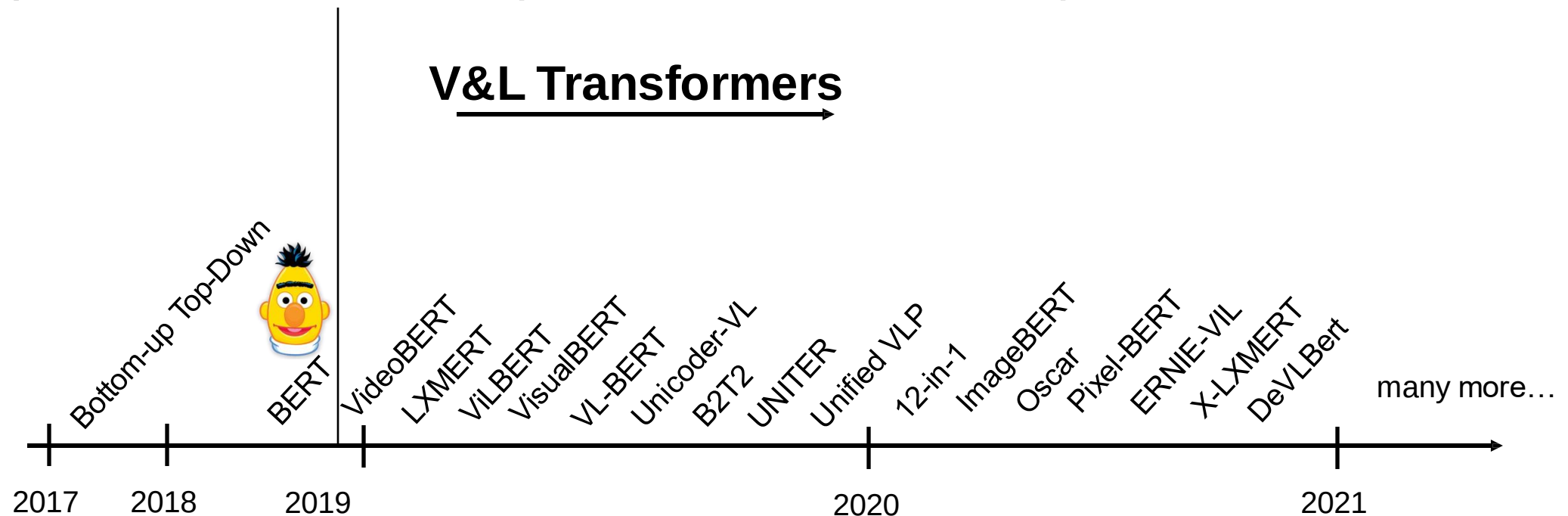


Visual Grounding

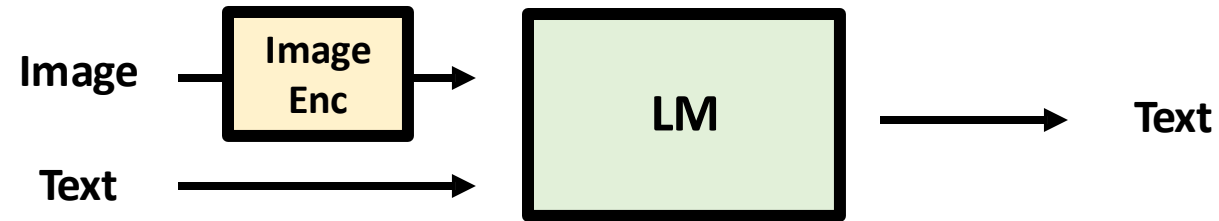


Vision-and-Language Pretraining Models

- Many different architectures (single vs. multi-stream), attention methods, objective functions, encoder/decoders, output heads, specialized modules (OCR/ASR/Tokenizers)

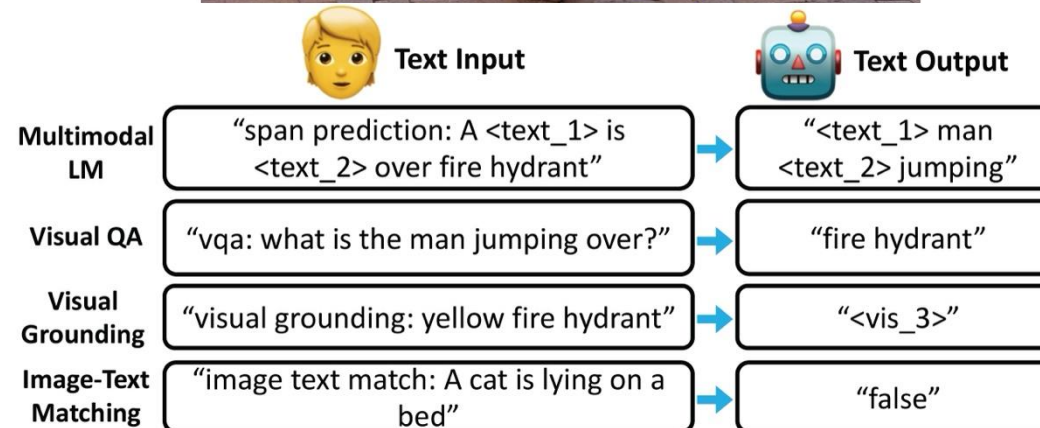
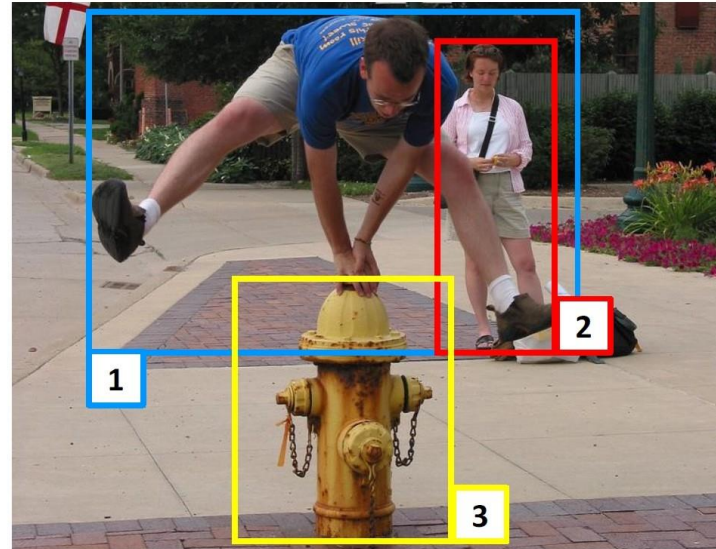


Vision-Language T5



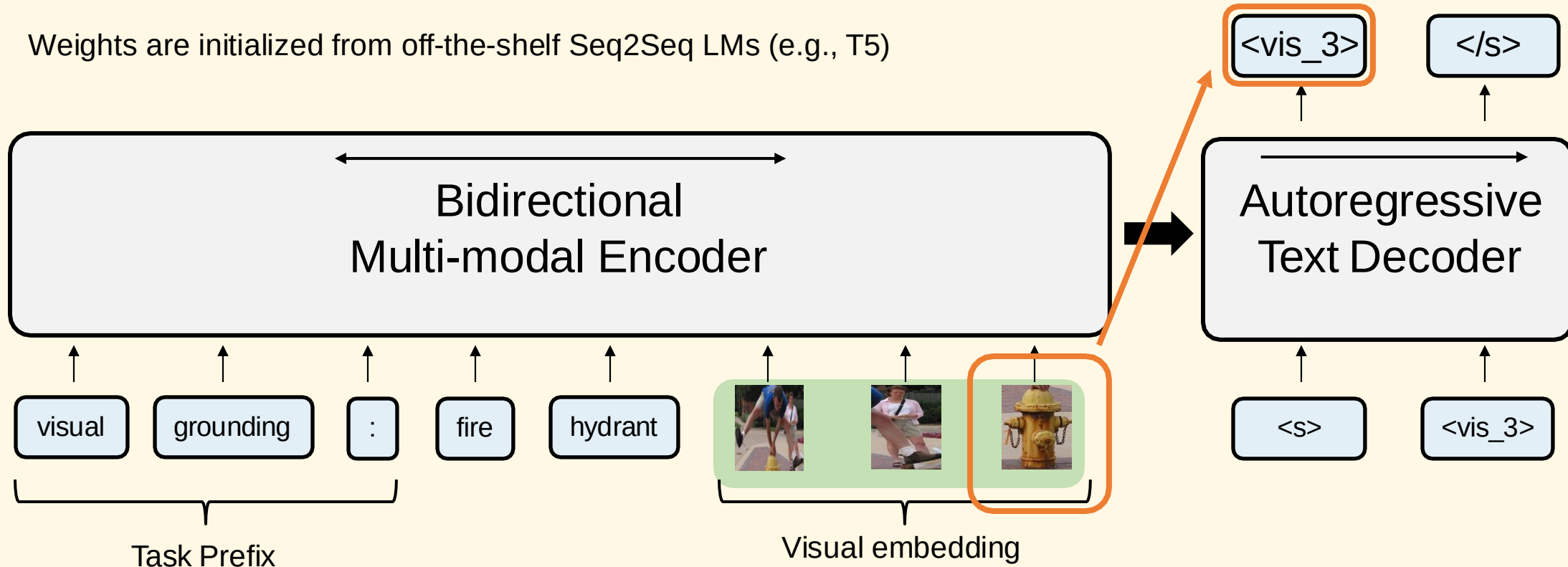
Can we tackle all V&L tasks with a single objective?

Many Multimodal Tasks as Text Generation

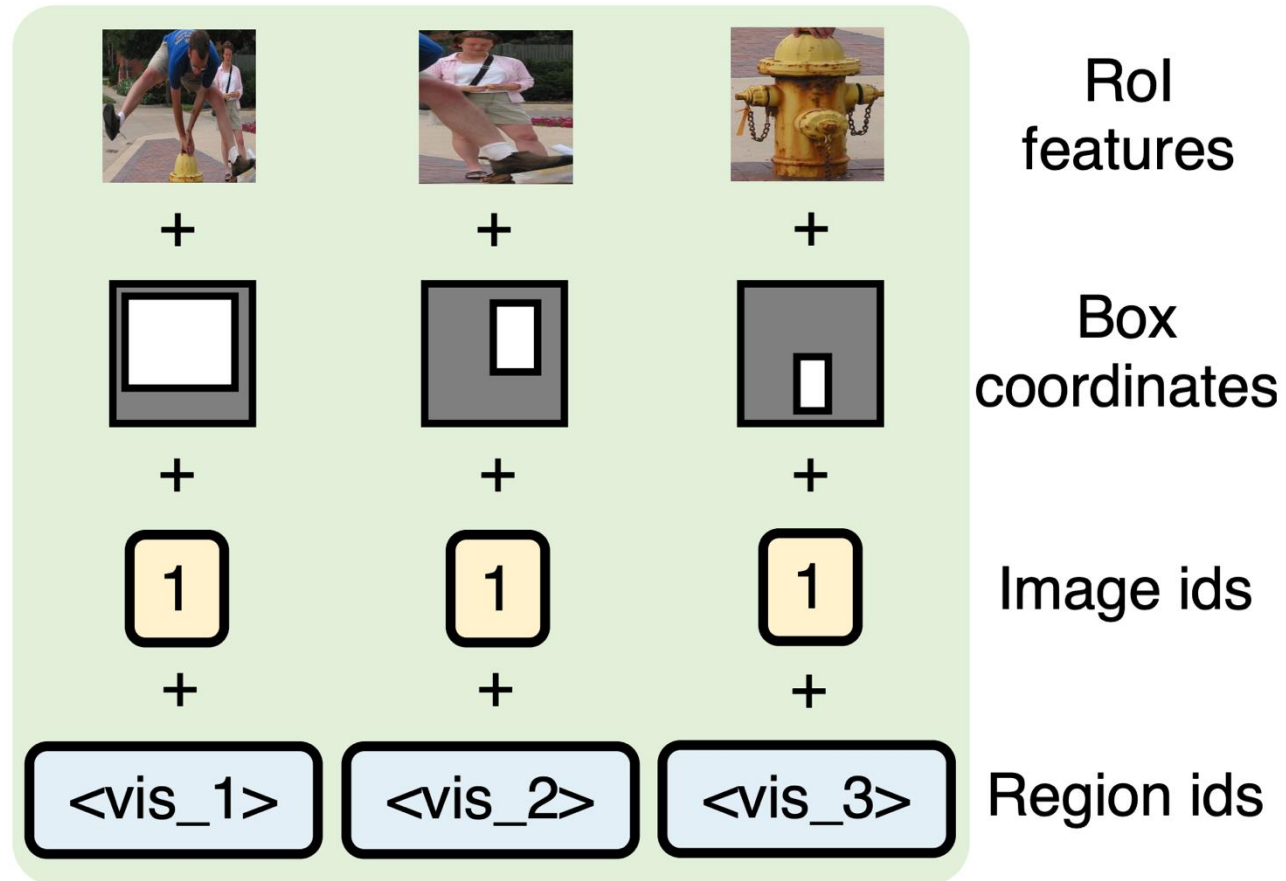


VL-T5 Architecture

Weights are initialized from off-the-shelf Seq2Seq LMs (e.g., T5)

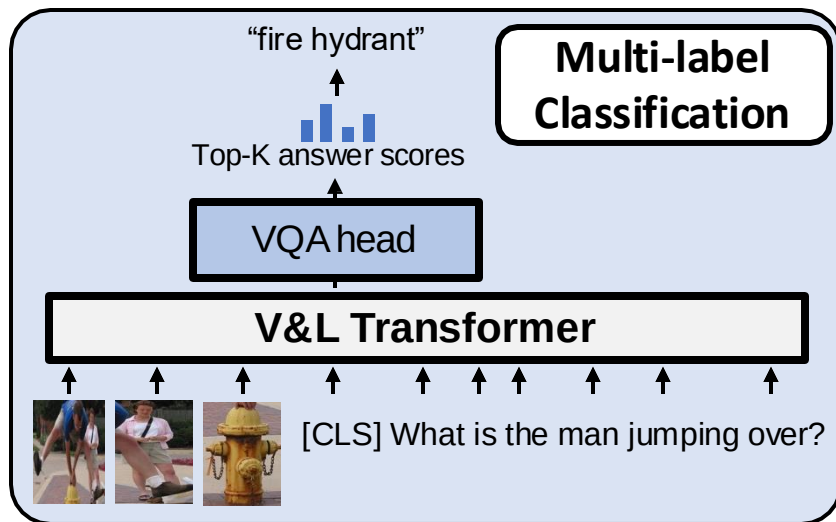


Visual embeddings

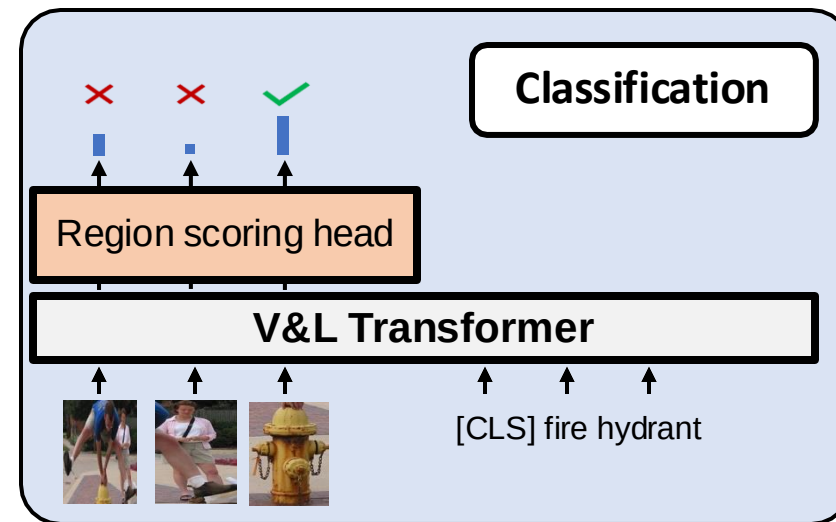


Previous models

Visual Question Answering

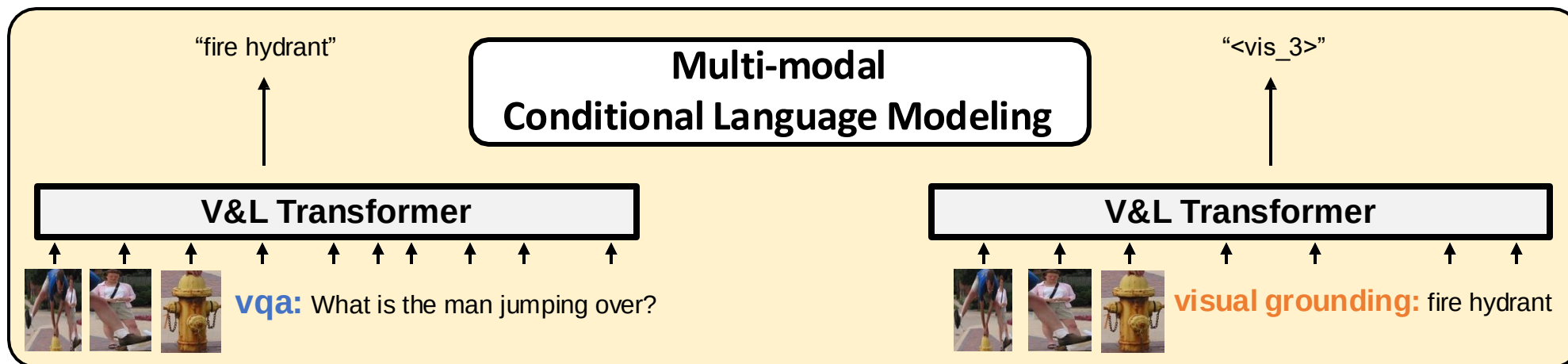


Visual Grounding



VL-T5

Multi-modal Conditional Language Modeling



Comparable to specialized models

Method	# Pretrain Images	Discriminative tasks					Generative tasks	
		VQA	GQA	NLVR ²	RefCOCOg	VCR Q→AR	COCO Cap	Multi30K En-De
		test-std Acc	test-std Acc	test-P Acc	test ^d Acc	test Acc	Karpathy test CIDEr	test 2018 BLEU
LXMERT	180K	72.5	60.3	74.5	-	-	-	-
ViLBERT	3M	70.9	-	-	-	54.8	-	-
UNITER _{Base}	4M	72.9	-	77.9	74.5	58.2	-	-
Unified VLP	3M	70.7	-	-	-	-	117.7	-
Oscar _{Base}	4M	73.4	61.6	78.4	-	-	123.7	-
XGPT	3M	-	-	-	-	-	120.1	-
MeMAD	-	-	-	-	-	-	-	38.5
VL-T5	180K	70.3	60.8	73.6	71.3	58.9	116.5	38.6
VL-BART	180K	71.3	60.5	70.3	22.4 [*]	48.9	116.6	28.1

Multi-task Learning with shared parameters

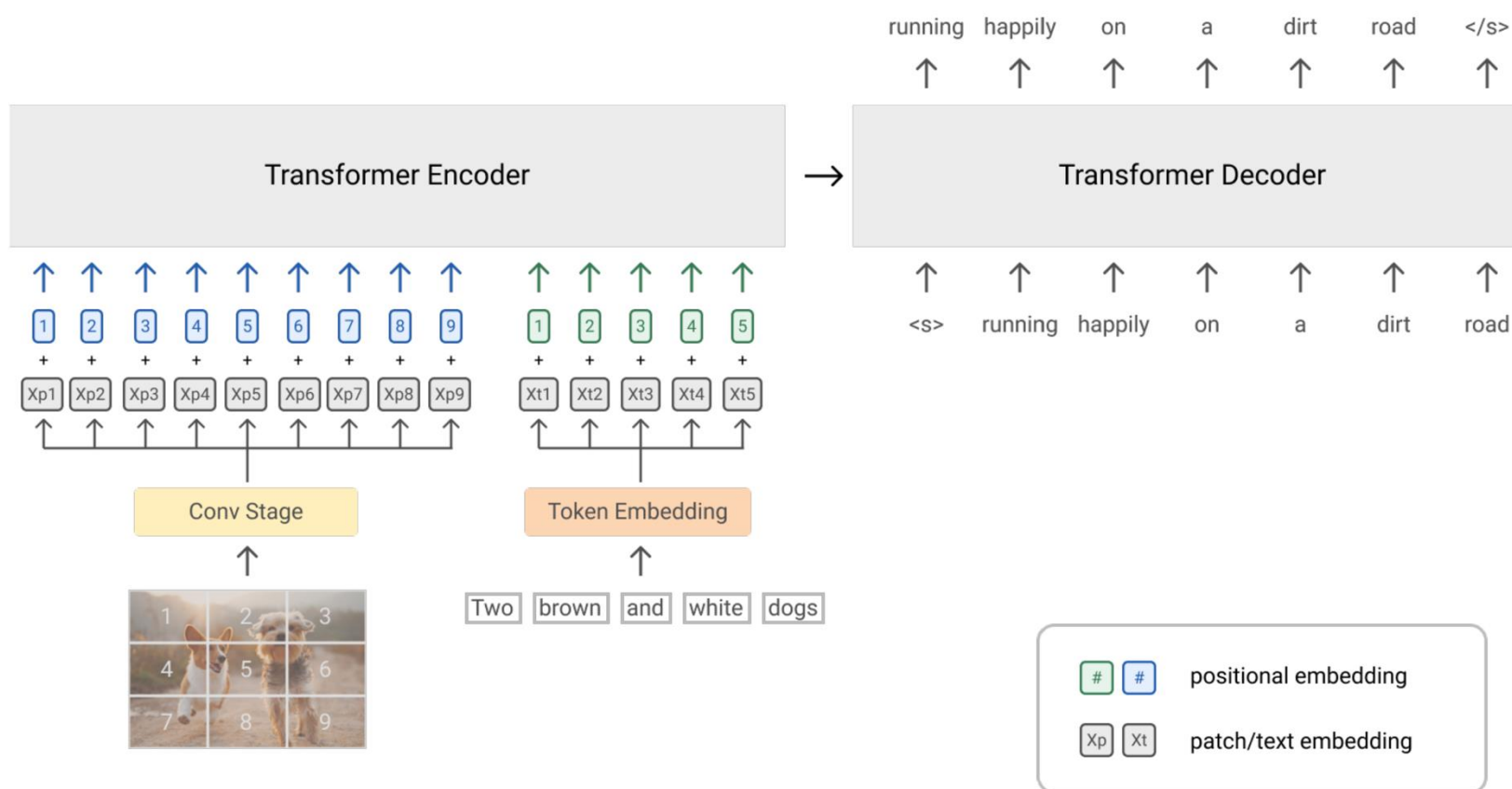
Method	Finetuning tasks	# Params	Discriminative tasks					Generative tasks	
			VQA Karpathy test Acc	GQA test-dev Acc	NLVR ² test-P Acc	RefCOCOg test ^d Acc	VCR val Acc	COCO Caption Karpathy test CIDEr	Multi30K En-De test2018 BLEU
VL-T5	single task	7P	67.9	60.0	73.6	71.3	57.5	116.1	38.6
VL-T5	all tasks	P	67.2	58.9	71.6	69.4	55.3	110.8	37.6

Similar performance with $1/7^{\text{th}} = 14\%$ parameters!

- Also performs better on rare/unseen categories!

SimVLM

- Train from scratch on a large-scale weakly-supervised

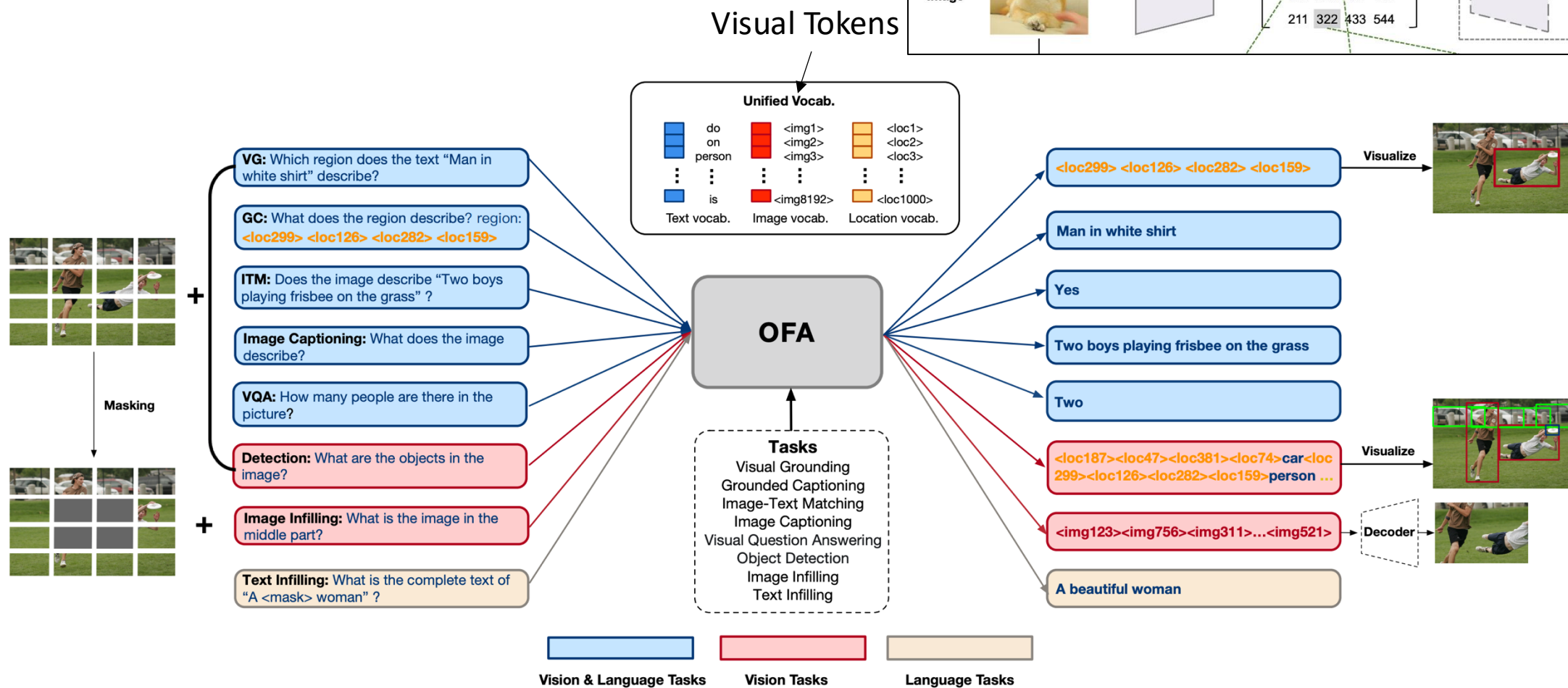


Pre-training data

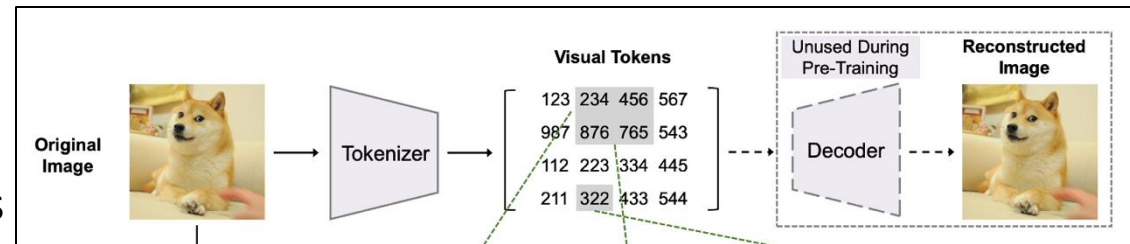
- Each batch has 4096 image-text pairs (ALIGN) and 512 text-only documents (C4)
- Both the image-text and text-only were important

Method	VQA score
No Pretraining	49.70
Decoder-only	65.23
w/ LM	64.48
SimVLM _{small}	67.43
w/o Image2Text	49.23
w/o Text2Text	65.25
w/o conv stage	63.11
w/ span corruption	66.23
w/ 2 conv blks	65.57
w/ 4 conv blks	66.55
w/ 10% ALIGN	66.71
w/ CC-3M	63.32

OFA



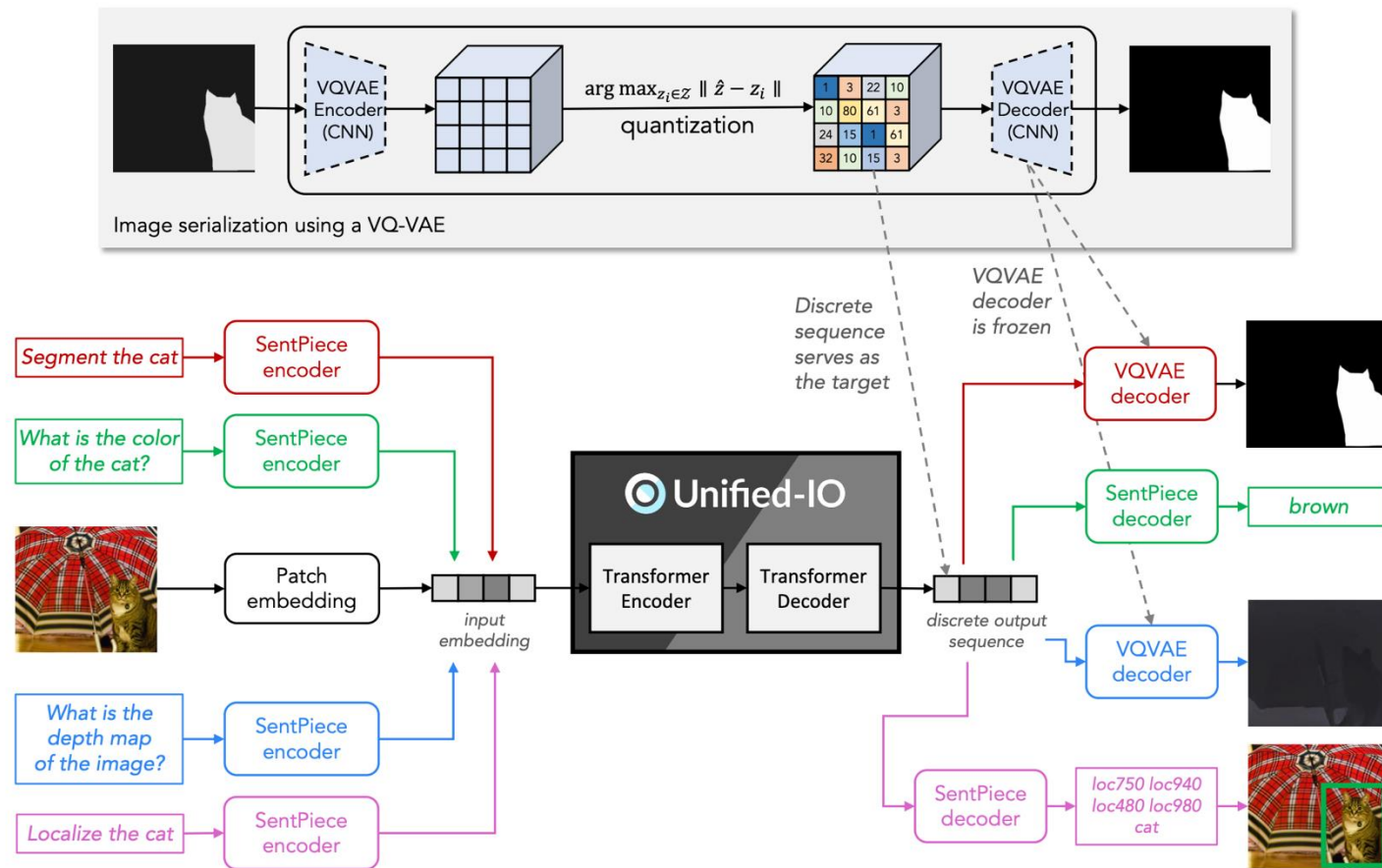
Recall:



Quantative results

Model	VQA		SNLI-VE	
	test-dev	test-std	dev	test
UNITER [14]	73.8	74.0	79.4	79.4
OSCAR [15]	73.6	73.8	-	-
VILLA [16]	74.7	74.9	80.2	80.0
VL-T5 [56]	-	70.3	-	-
VinVL [17]	76.5	76.6	-	-
UNIMO [46]	75.0	75.3	81.1	80.6
ALBEF [69]	75.8	76.0	80.8	80.9
METER [70]	77.7	77.6	80.9	81.2
VLMo [48]	79.9	80.0	-	-
SimVLM [22]	80.0	80.3	86.2	86.3
Florence [23]	80.2	80.4	-	-
OFA _{Tiny}	70.3	70.4	85.3	85.2
OFA _{Medium}	75.4	75.5	86.6	87.0
OFA _{Base}	78.0	78.1	89.3	89.2
OFA _{Large}	80.3	80.5	90.3	90.2
OFA	82.0	82.0	91.0	91.2

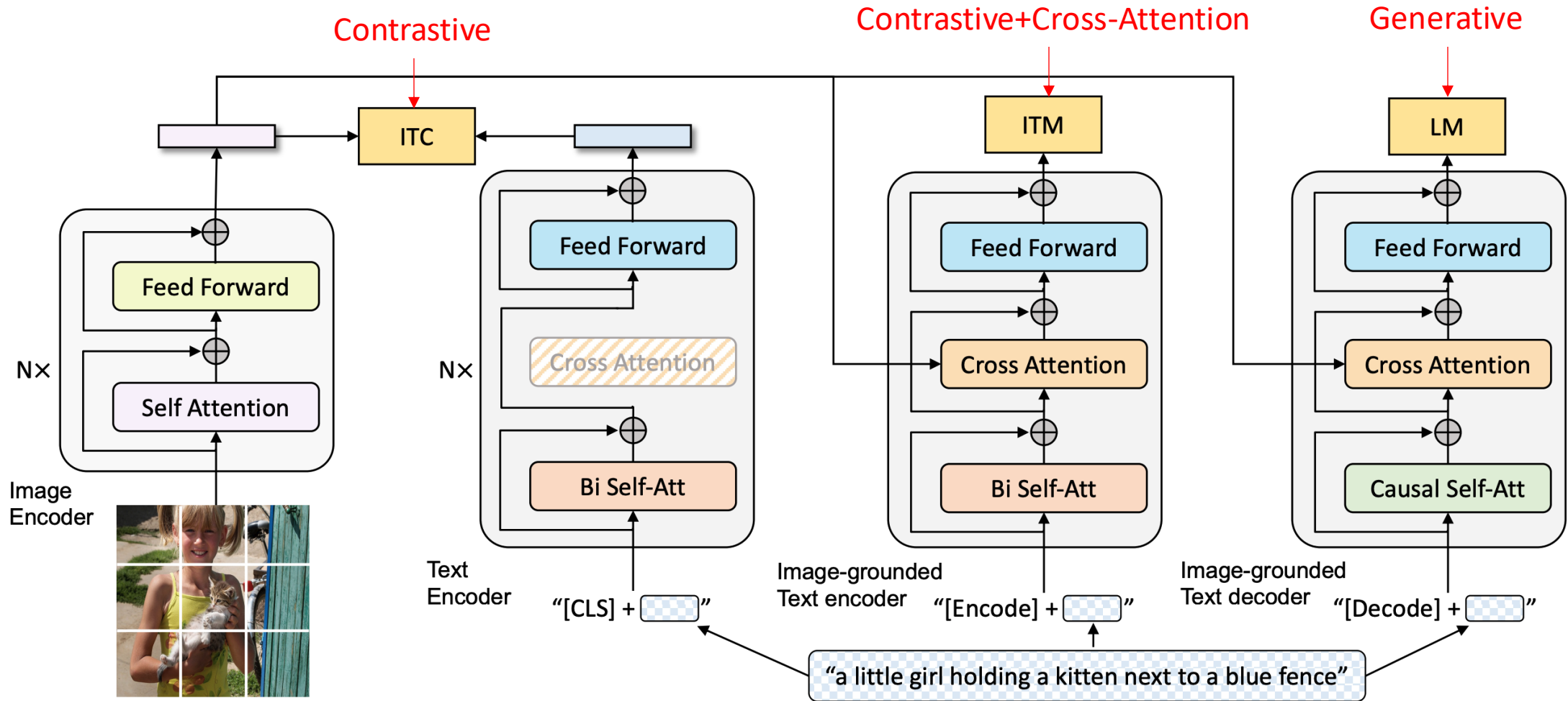
Unified-IO (concurrent work)



Quantitative results

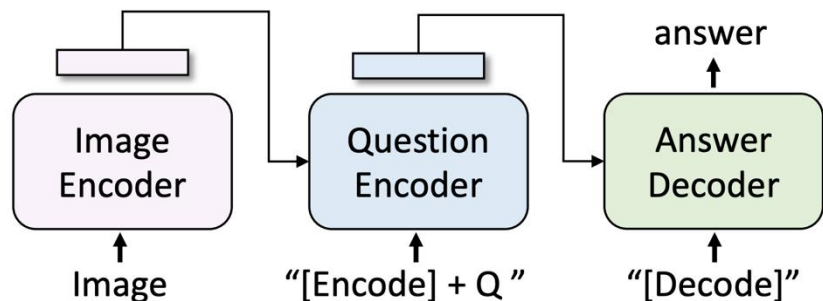
		Categorization		Localization		VQA		Refexp		Segmentation		Keypoint		Normal		All	
		ablation	test	ablation	test	ablation	test	ablation	test	ablation	test	ablation	test	ablation	test	ablation	test
0	NLL-AngMF [4]	-	-	-	-	-	-	-	-	-	-	-	-	49.6	50.5	7.2	7.1
1	Mask R-CNN [41]	-	-	44.7	45.1	-	-	-	-	26.2	26.2	70.8	70.6	-	-	20.2	20.3
2	GPV-1 [38]	33.2	33.2	42.8	42.7	50.6	49.8	25.8	26.8	-	-	-	-	-	-	21.8	21.8
3	CLIP [86]	48.1	-	-	-	-	-	-	-	-	-	-	-	-	-	6.9	-
4	OFA _{LARGE} [107]	22.6	-	-	-	72.4	-	61.7	-	-	-	-	-	-	-	22.4	-
5	GPV-2 [52]	54.7	55.1	53.6	53.6	63.5	63.2	51.5	52.1	-	-	-	-	-	-	31.9	32.0
6	UNIFIED-IO _{SMALL}	42.6	-	50.4	-	52.9	-	51.1	-	40.7	-	46.5	-	33.5	-	45.4	-
7	UNIFIED-IO _{BASE}	53.1	-	59.7	-	63.0	-	68.3	-	49.3	-	60.2	-	37.5	-	55.9	-
8	UNIFIED-IO _{LARGE}	57.0	-	64.2	-	67.4	-	74.1	-	54.0	-	67.6	-	40.2	-	60.7	-
9	UNIFIED-IO _{XL}	61.7	60.8	67.0	67.1	74.5	74.5	78.6	78.9	56.3	56.5	68.1	67.7	45.0	44.3	64.5	64.3

BLIP: Unified Understanding and Generation

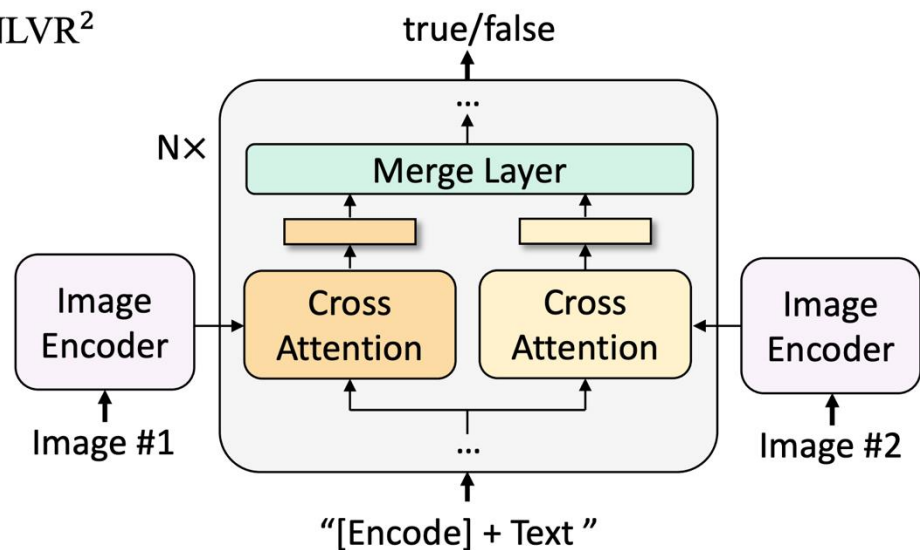


BLIP results

(a) VQA

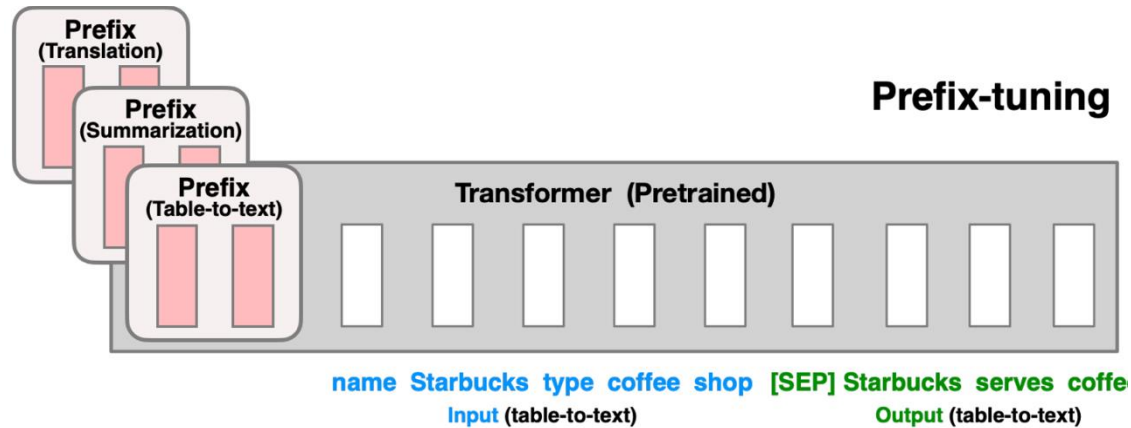


(b) NLVR²

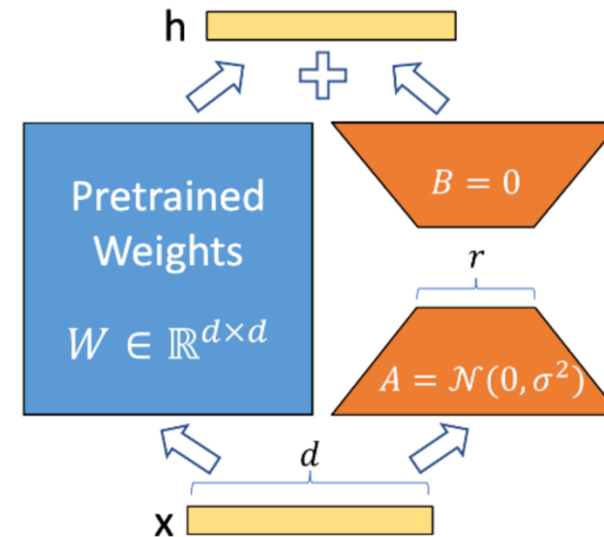


Method	Pre-train #Images	VQA		NLVR ²	
		test-dev	test-std	dev	test-P
LXMERT	180K	72.42	72.54	74.90	74.50
UNITER	4M	72.70	72.91	77.18	77.85
VL-T5/BART	180K	-	71.3	-	73.6
OSCAR	4M	73.16	73.44	78.07	78.36
SOHO	219K	73.25	73.47	76.37	77.32
VILLA	4M	73.59	73.67	78.39	79.30
UNIMO	5.6M	75.06	75.27	-	-
ALBEF	14M	75.84	76.04	82.55	83.14
SimVLM _{base} [†]	1.8B	77.87	78.14	81.72	81.77
BLIP	14M	77.54	77.62	82.67	82.30
BLIP	129M	78.24	78.17	82.48	83.08
BLIP _{CapFilt-L}	129M	78.25	78.32	82.15	82.24

Parameter Efficient Fine-tuning



Prefix-tuning



Low-rank adaption (LoRA)

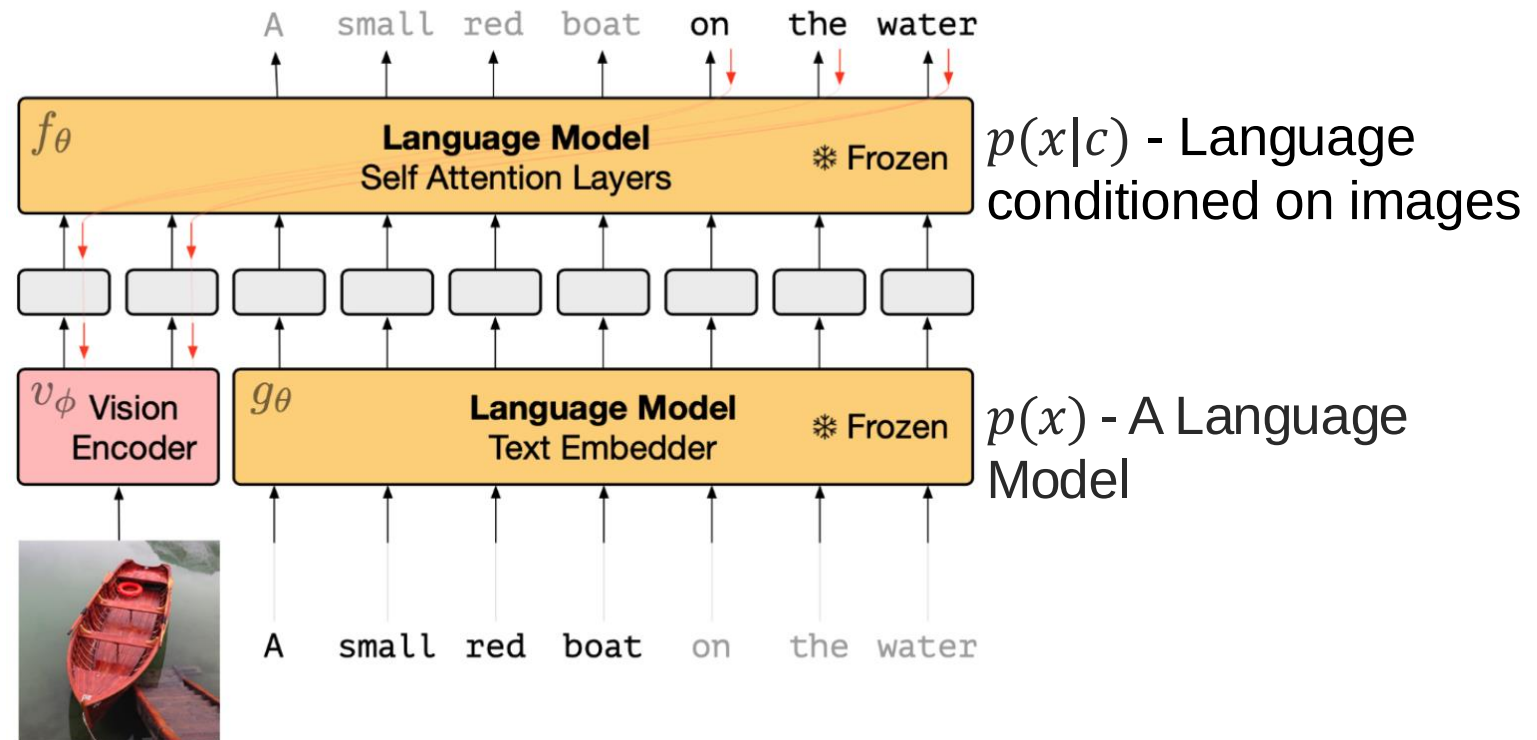
About 99% of the model kept frozen

Prefix-Tuning: Optimizing Continuous Prompts for Generation; Li et al.

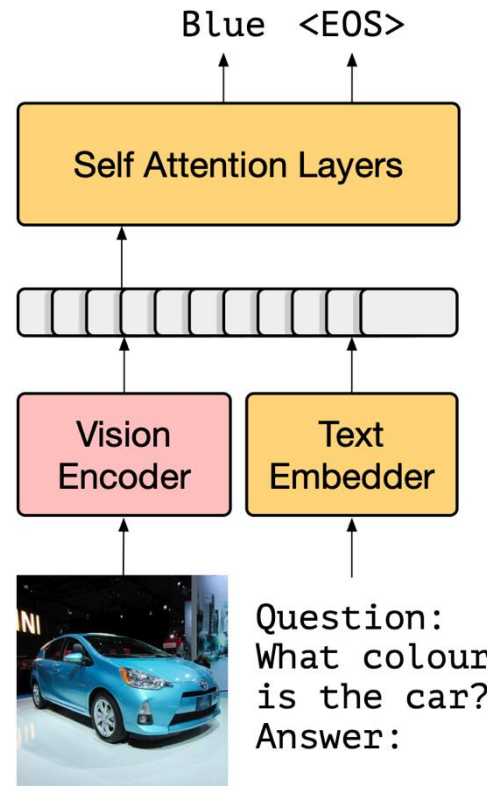
LoRA: Low-Rank Adaptation of Large Language Models; Hu et al.

Conditioning with frozen models

Trained with
image captioning
datasets



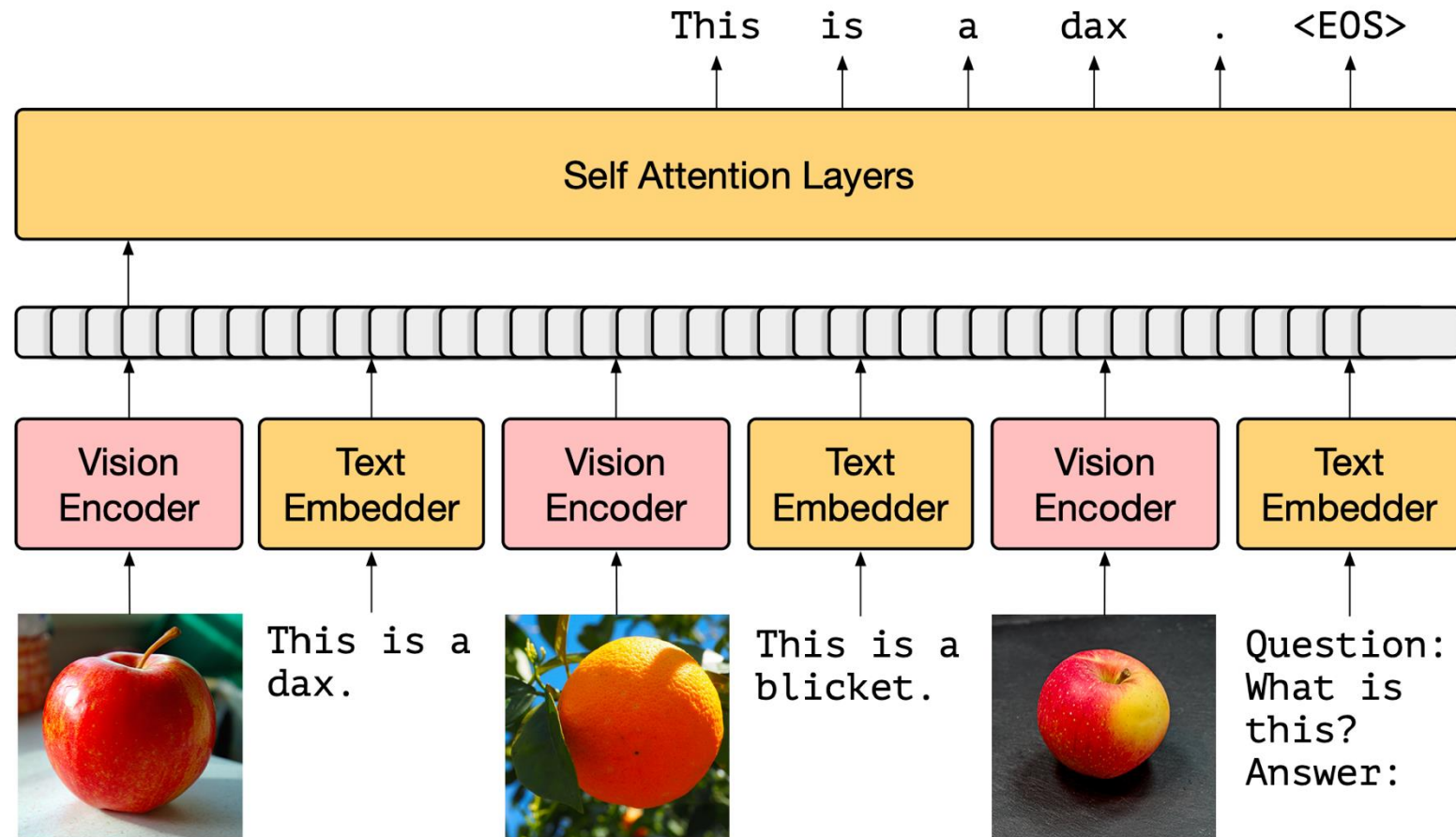
0-shot VQA



$p(x|c)$ - Language
conditioned on images

$p(x)$ - A Language
Model

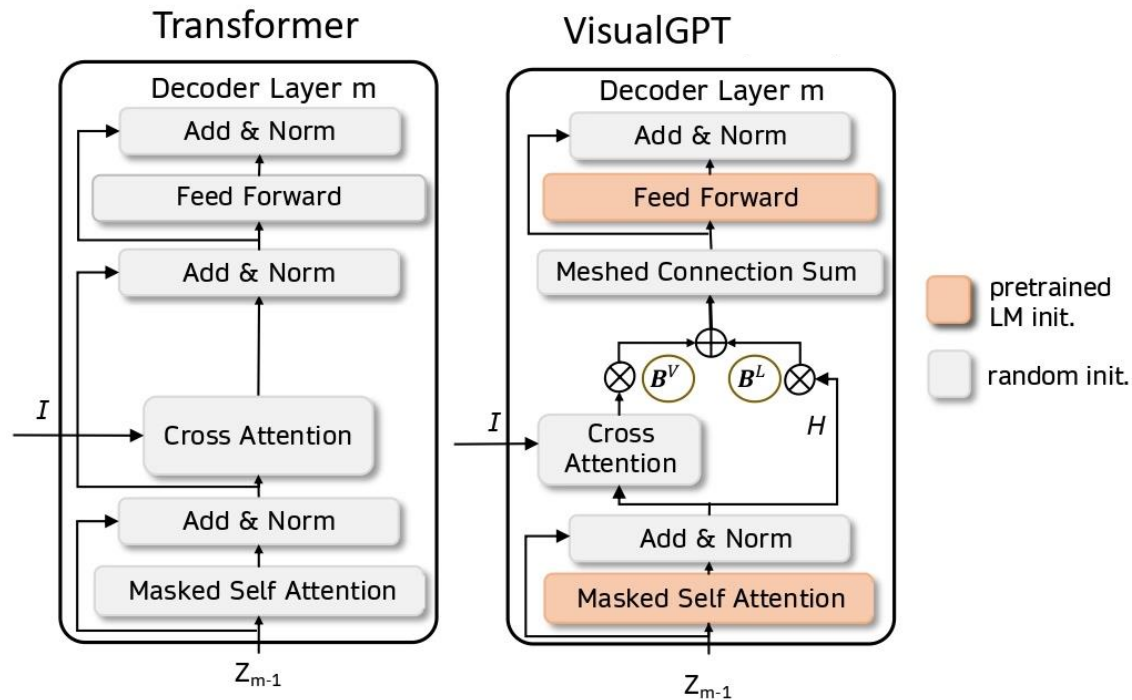
Few-shot image classification



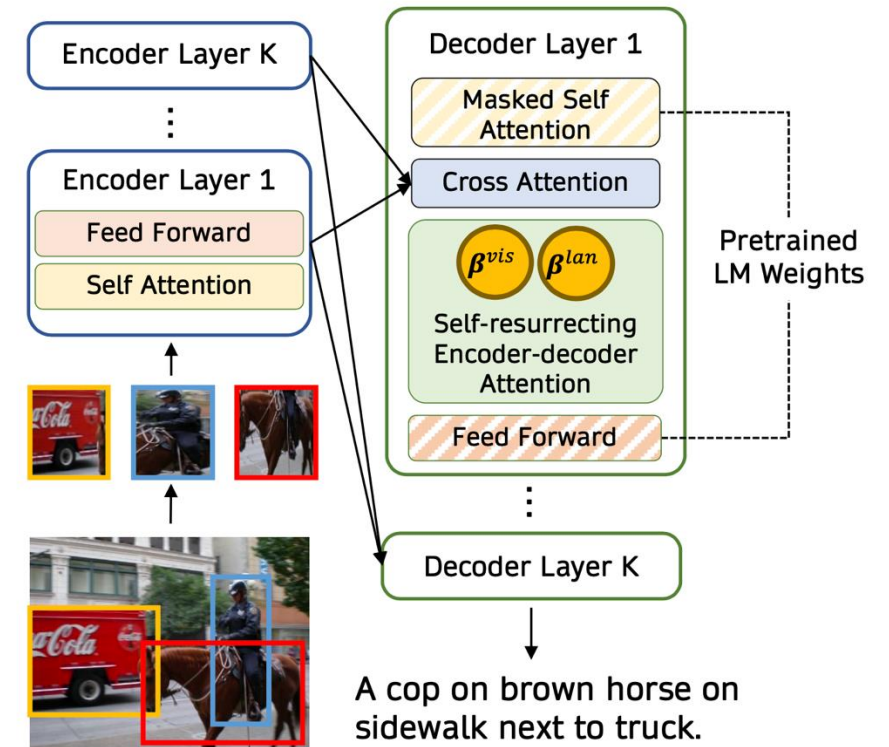
Transfer-learning from Captions to VQA

Text-Image Cross-Attention Fuse Mechanisms

- Cross-attention

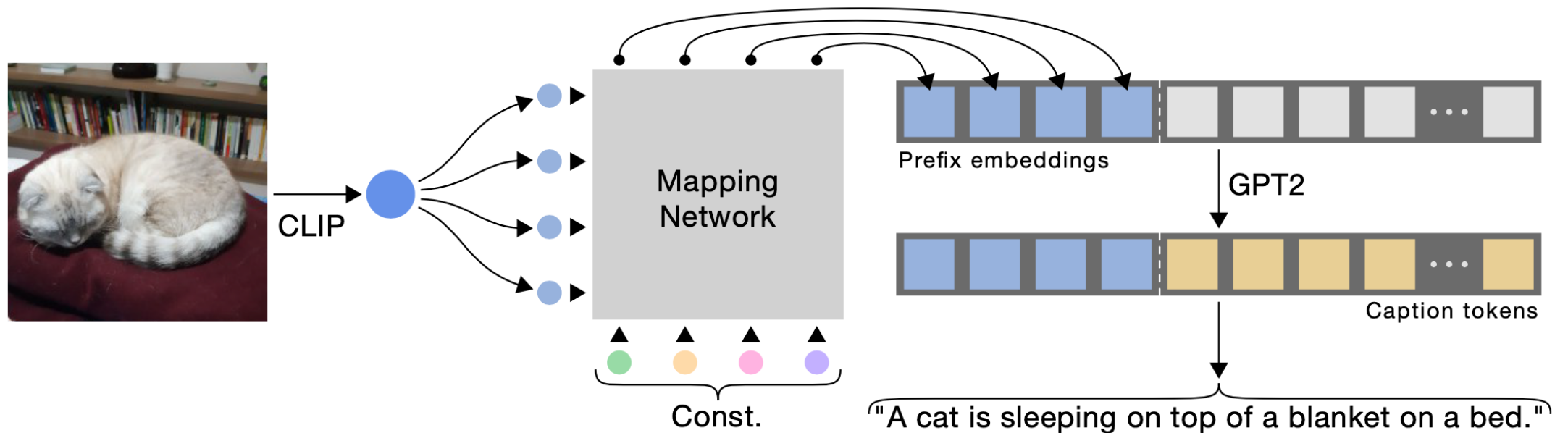


Detailed sketch:



ClipCap: Mapping from CLIP to GPT

- Both CLIP and GPT2 are frozen

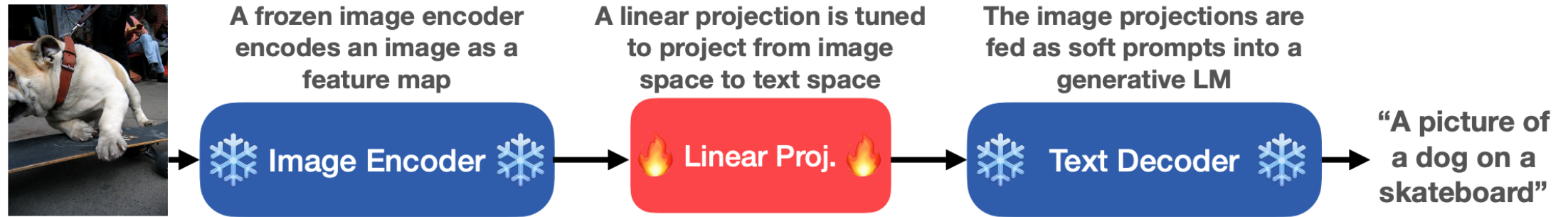


Employing frozen networks is efficient

(A) **Conceptual Captions**

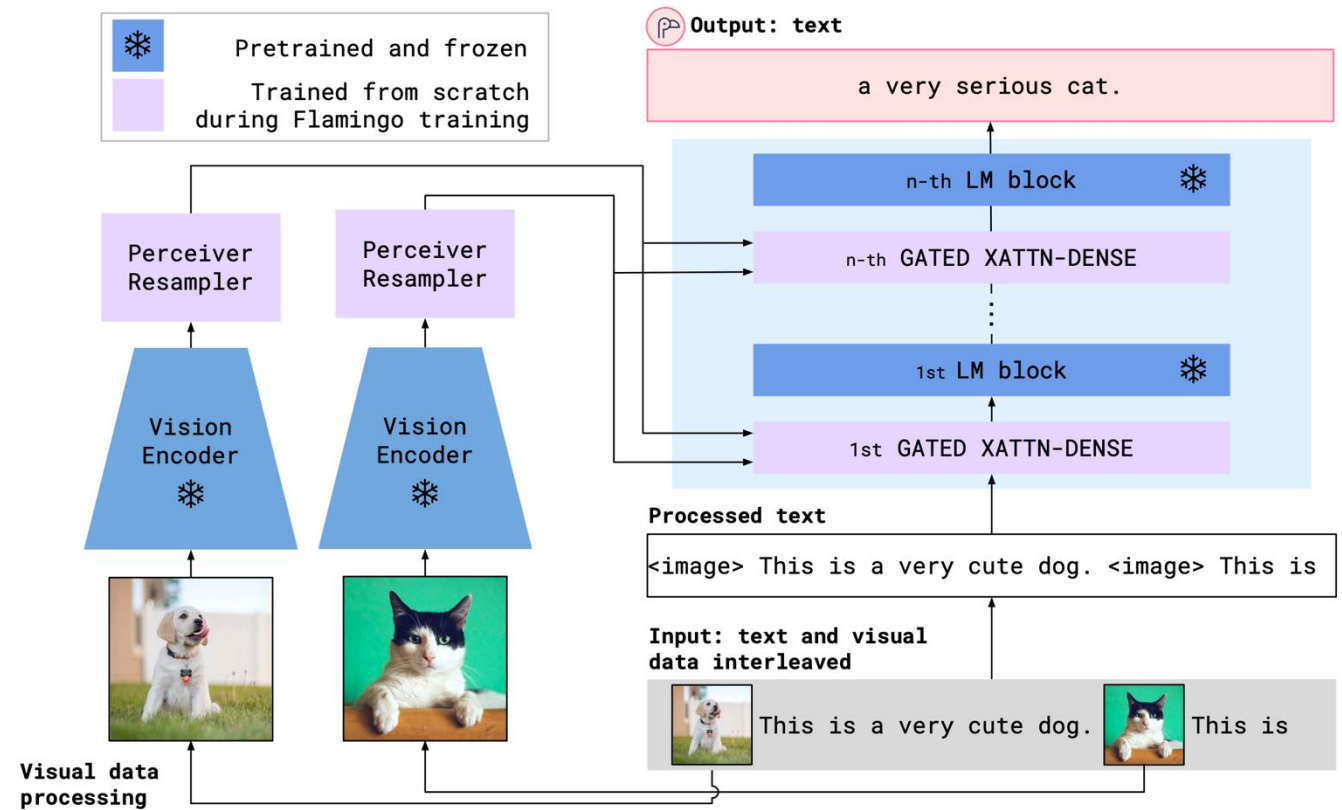
Model	ROUGE-L $\uparrow$	CIDEr $\uparrow$	SPICE $\uparrow$	#Params (M) $\downarrow$	Training Time $\downarrow$
VLP	24.35	77.57	16.59	115	1200h (V100)
Ours; MLP + GPT2 tuning	26.71	87.26	18.5	156	80h (GTX1080)
Ours; Transformer	25.12	71.82	16.07	43	72h (GTX1080)

Linear mapping from image to text



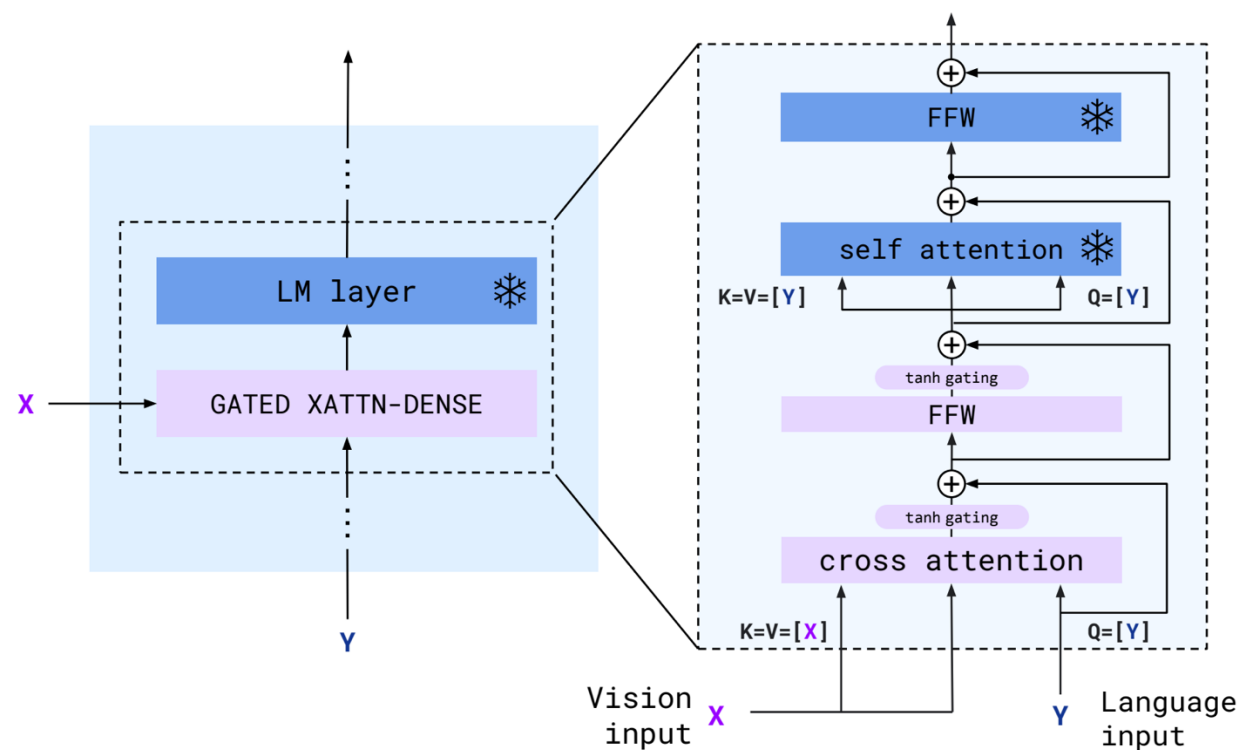
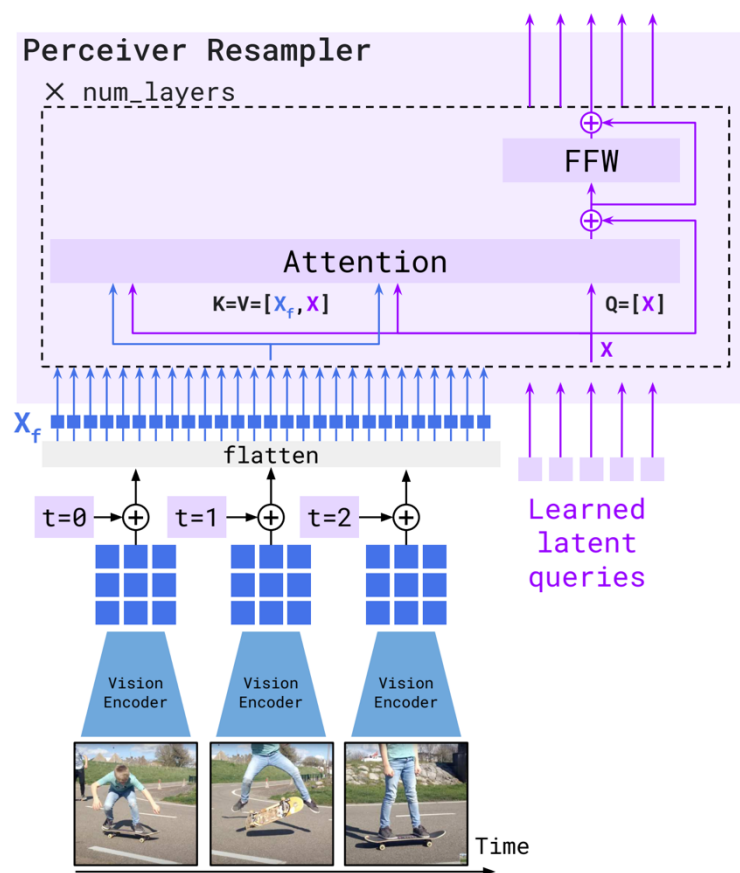
Flamingo

- Employ 70B Frozen LLM
- Large-scale dataset: 43M web pages, Multimodal MassiveWeb (M3W)

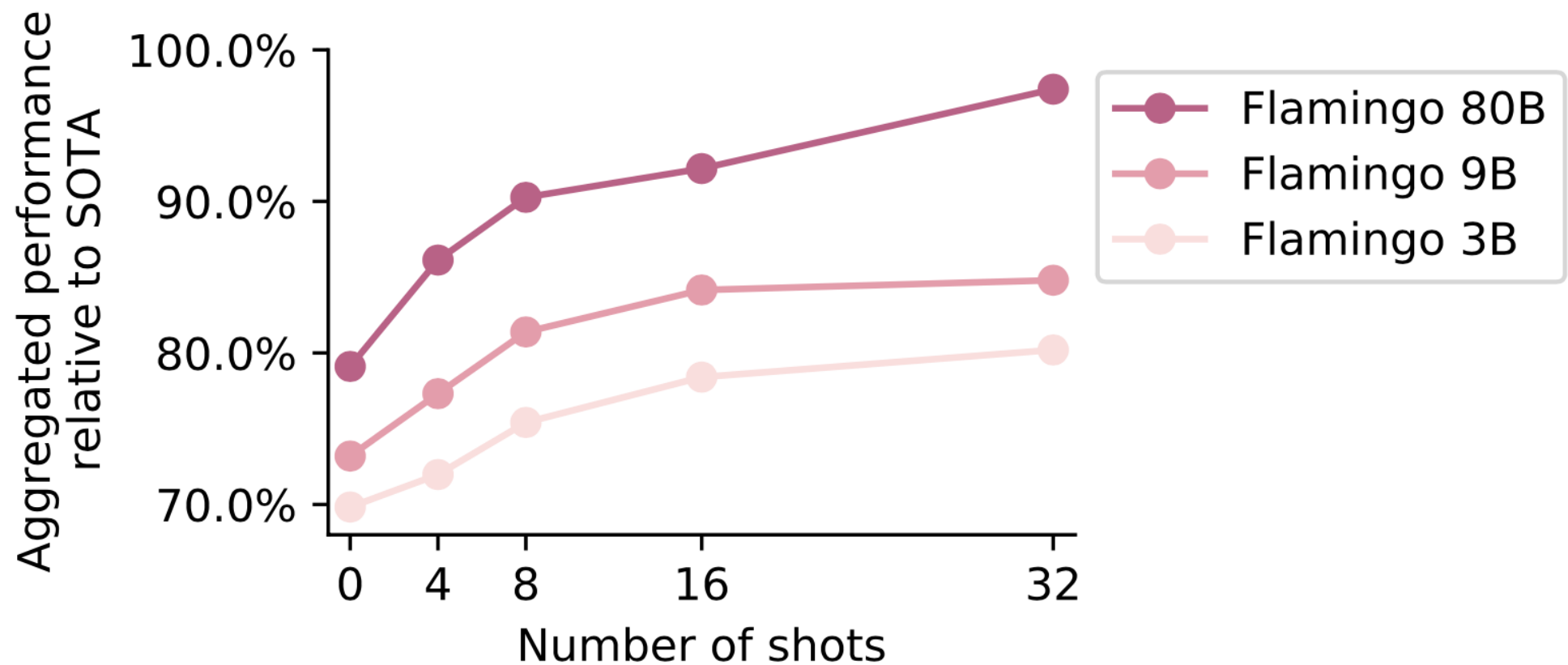


Perceiver

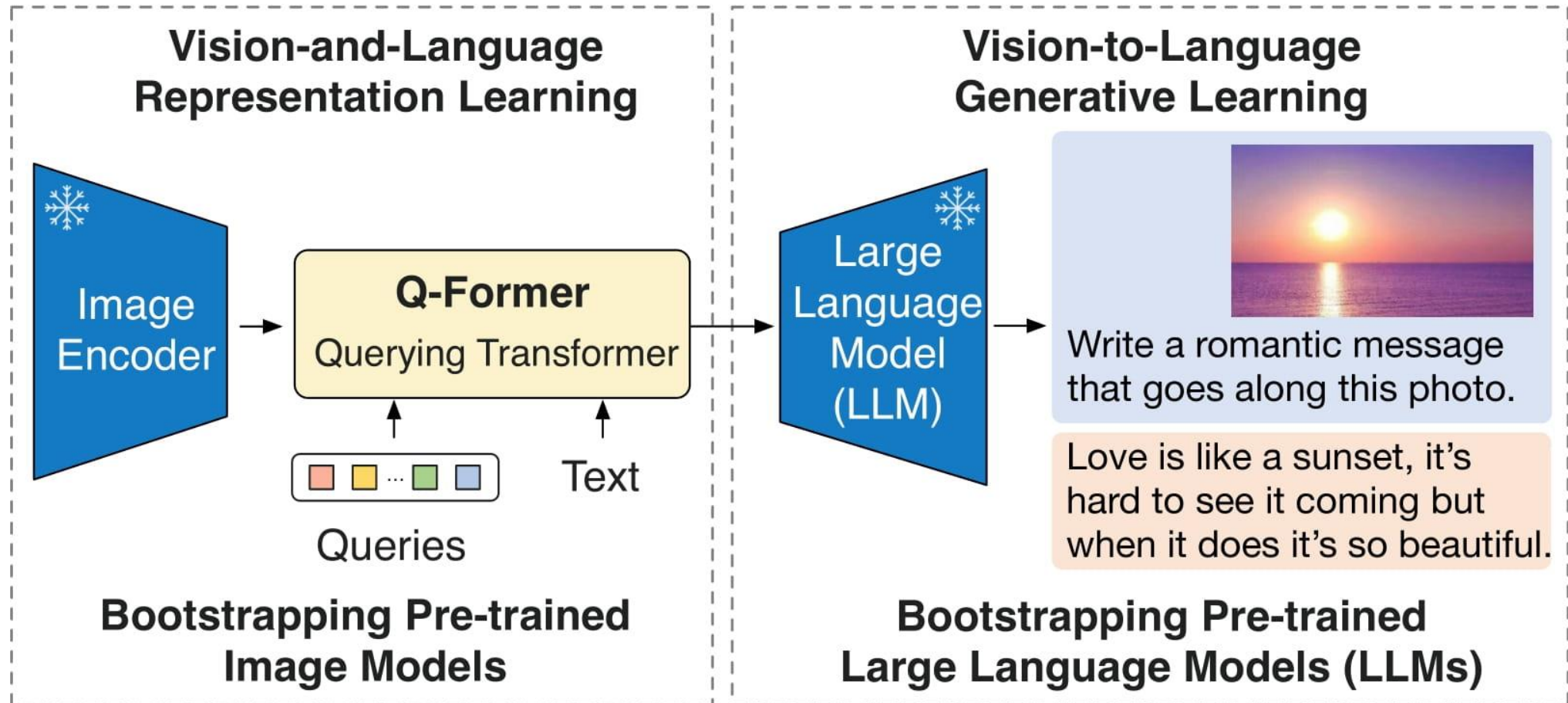
- Perceiver-based architecture produces a fixed number of tokens



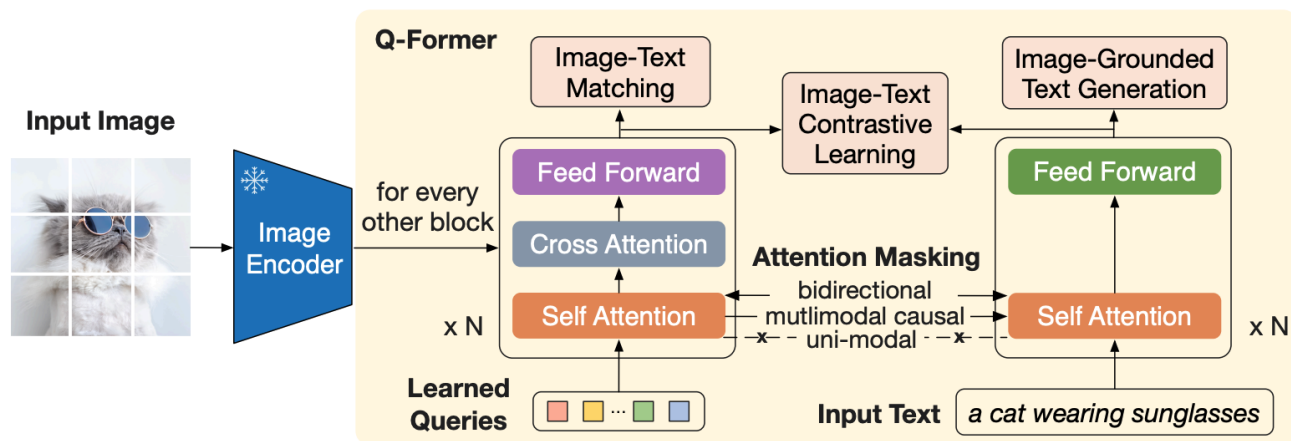
Few-shot learning



BLIP-2: Frozen Image Encoders

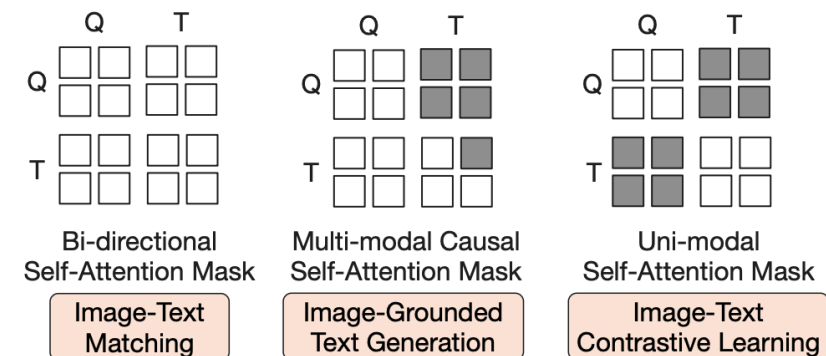


Q-Former

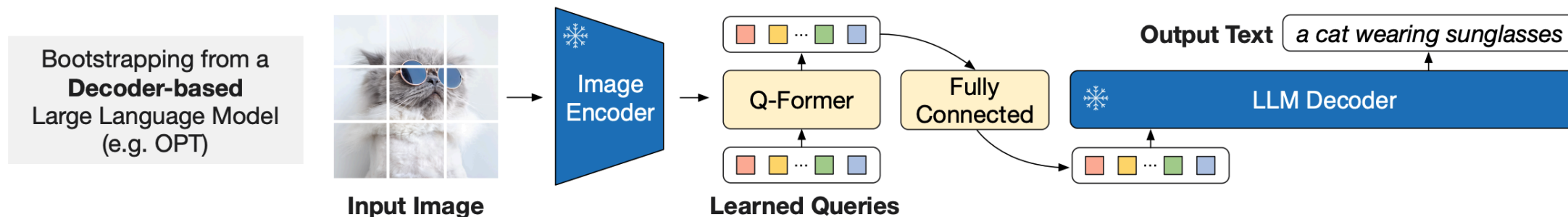


Q: query token positions; **T:** text token positions.

■ masked □ unmasked

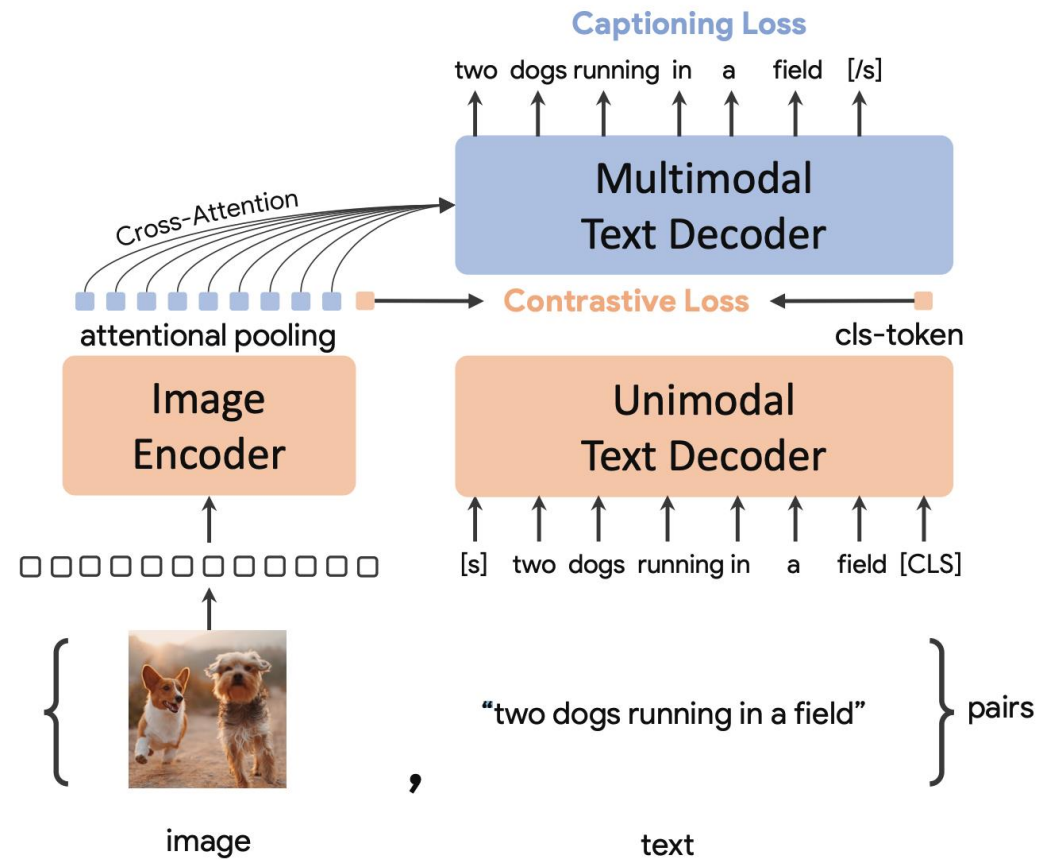


Quantative results

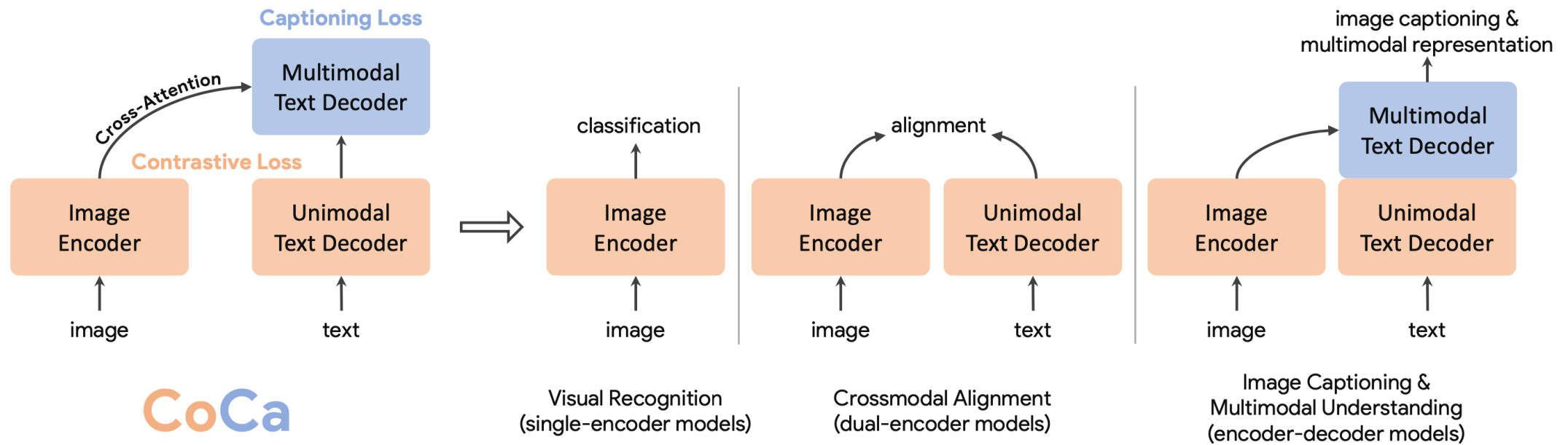


Models	#Trainable Params	VQAv2	
		test-dev	test-std
<i>Open-ended generation models</i>			
ALBEF (Li et al., 2021)	314M	75.84	76.04
BLIP (Li et al., 2022)	385M	78.25	78.32
OFA (Wang et al., 2022a)	930M	82.00	82.00
Flamingo80B (Alayrac et al., 2022)	10.6B	82.00	82.10
BLIP-2 ViT-g FlanT5 _{XL}	1.2B	81.55	81.66
BLIP-2 ViT-g OPT _{2.7B}	1.2B	81.59	81.74
BLIP-2 ViT-g OPT _{6.7B}	1.2B	82.19	82.30

CoCa: Contrastive Captioners



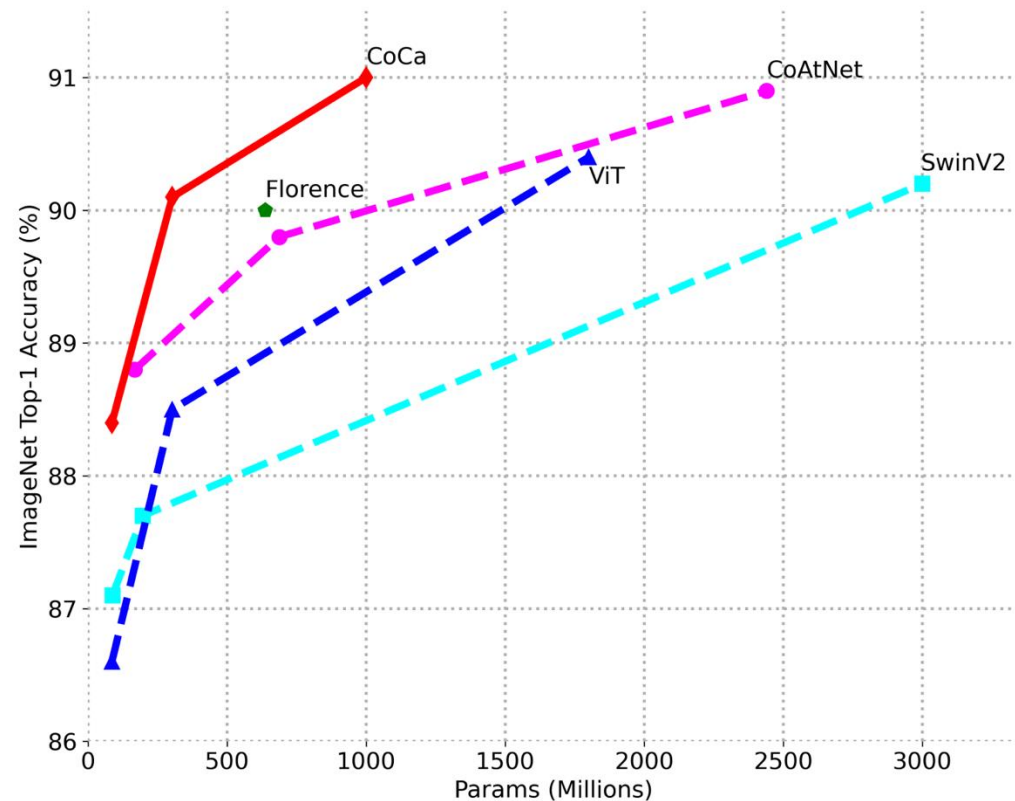
Can solve various downstream tasks



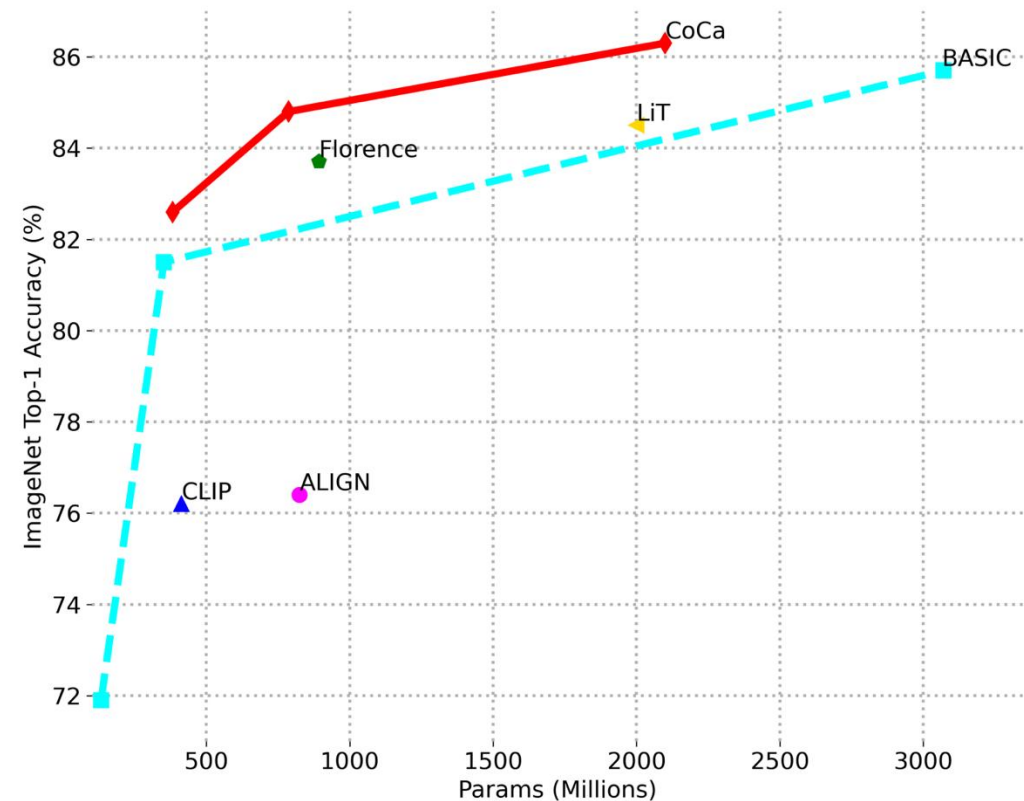
CoCa
Pretraining

Zero-shot, frozen-feature or finetuning

Great representations and zero-shot performance



(a) Finetuned ImageNet Top-1 Accuracy.



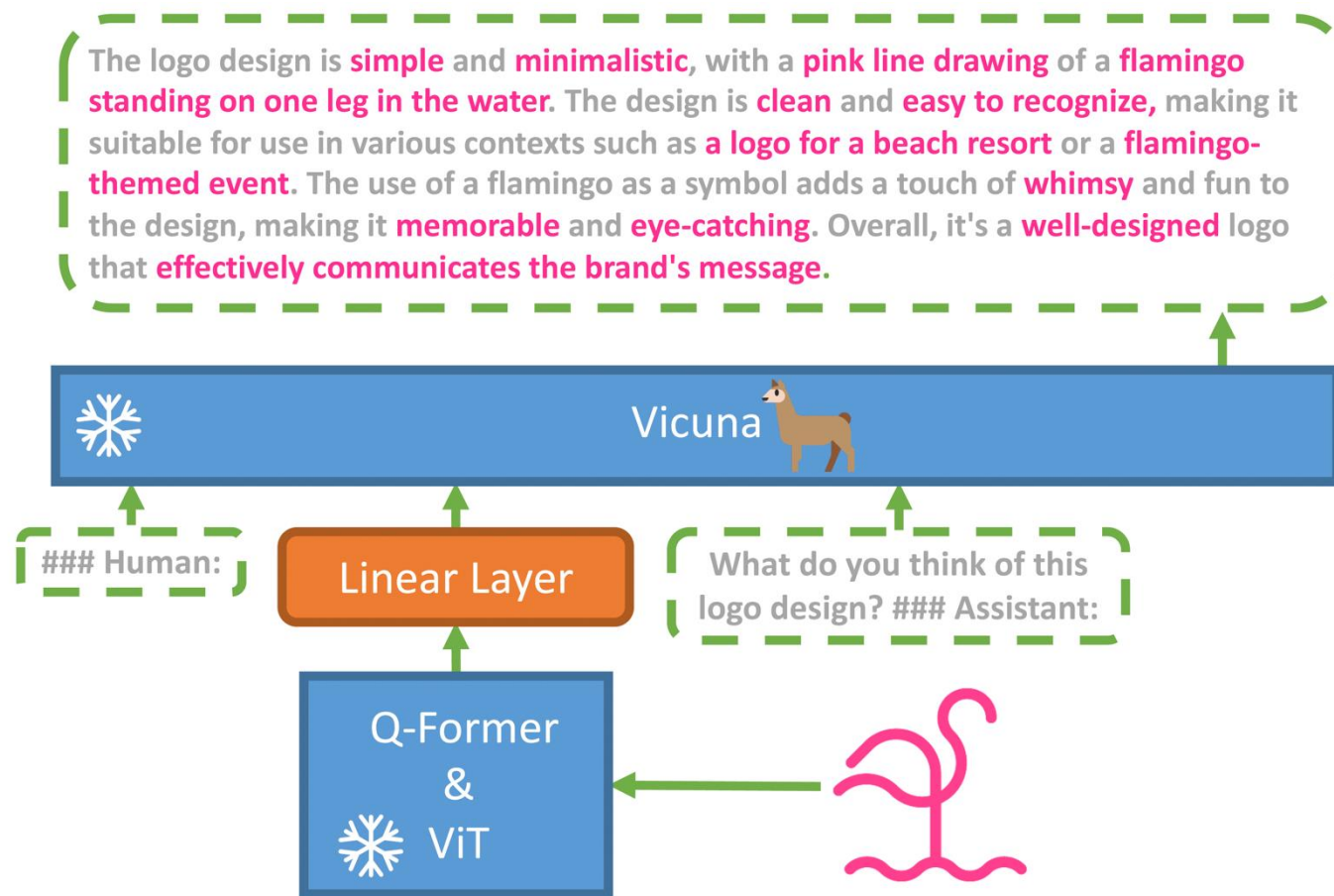
(b) Zero-Shot ImageNet Top-1 Accuracy.

Quantative results

Model	VQA		SNLI-VE		NLVR2	
	test-dev	test-std	dev	test	dev	test-p
UNITER [26]	73.8	74.0	79.4	79.4	79.1	80.0
VinVL [27]	76.6	76.6	-	-	82.7	84.0
CLIP-ViL [73]	76.5	76.7	80.6	80.2	-	-
ALBEF [36]	75.8	76.0	80.8	80.9	82.6	83.1
BLIP [37]	78.3	78.3	-	-	82.2	82.2
OFA [17]	79.9	80.0	90.3 [†]	90.2 [†]	-	-
VLMo [30]	79.9	80.0	-	-	85.6	86.9
SimVLM [16]	80.0	80.3	86.2	86.3	84.5	85.2
Florence [14]	80.2	80.4	-	-	-	-
METER [74]	80.3	80.5	-	-	-	-
CoCa	82.3	82.3	87.0	87.1	86.1	87.0

Mini-GPT-4

- Stronger LLM (Vicuna)
- Instruction tuning



LLaVA: Visual Instruction Data

Context type 1: Captions

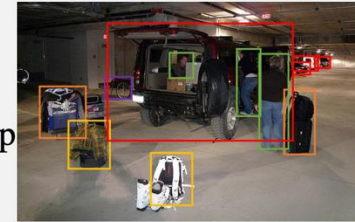
A group of people standing outside of a black vehicle with various luggage.

Luggage surrounds a vehicle in an underground parking area

People try to fit all of their luggage in an SUV.

The sport utility vehicle is parked in the public garage, being packed for a trip

Some people with luggage near a van that is transporting it.



Context type 2: Boxes

person: [0.681, 0.242, 0.774, 0.694], backpack: [0.384, 0.696, 0.485, 0.914], suitcase: ...<omitted>

Response type 1: conversation

Question: What type of vehicle is featured in the image?

Answer: The image features a black sport utility vehicle (SUV) ...<omitted>

Response type 2: detailed description

The image is an underground parking area with a black sport utility vehicle (SUV) parked. There are three people in the scene, with one person standing closer to the left side of the vehicle, another person in the middle, and the third person on the right side. They are all working together to pack their luggage into the SUV for a trip. ...<omitted>

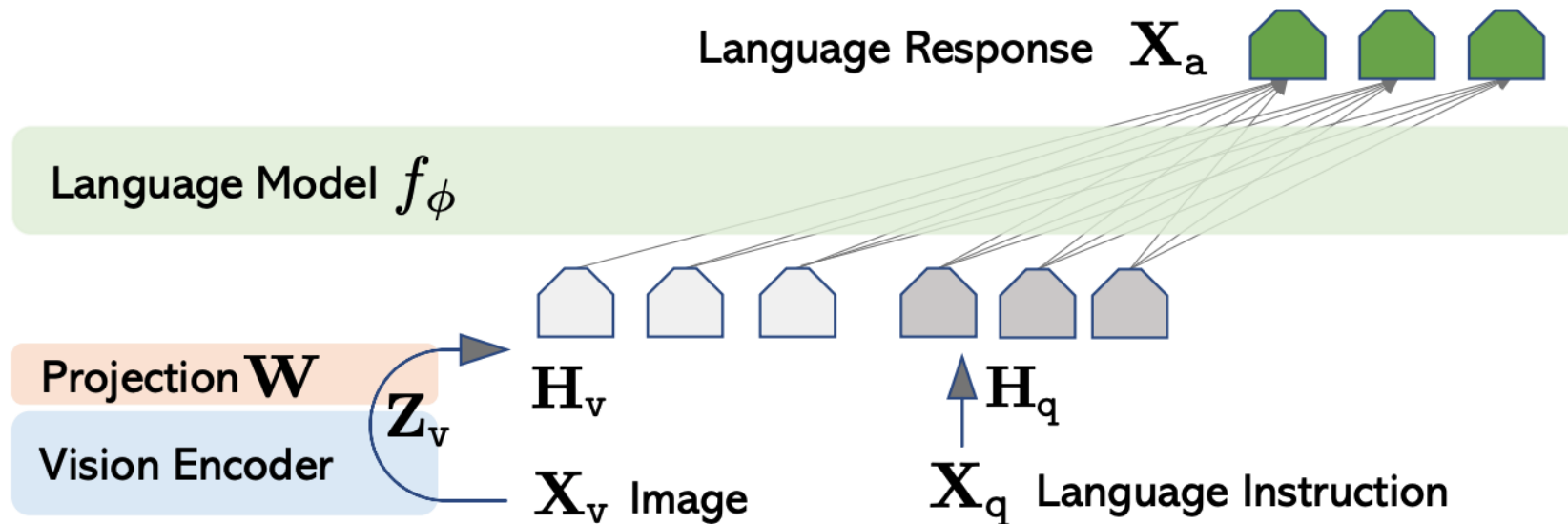
Response type 3: complex reasoning

Question: What challenges do these people face?

Answer: In the image, a group of people is standing outside a black SUV in a parking area, surrounded by various pieces of luggage, including suitcases and backpacks. They are facing the challenge of fitting all their luggage into the black SUV. There are multiple suitcases and backpacks to be packed, which suggests that the group has a significant amount of belongings ...<omitted>

Visual Instruction Tuning

- Conditioning with simple linear adaptars



Qualitative Example



What is unusual about this image?

Responses

BLIP-2: a man is sitting on the back of a yellow cab

OpenFlamingo: The man is drying his clothes on the hood of his car

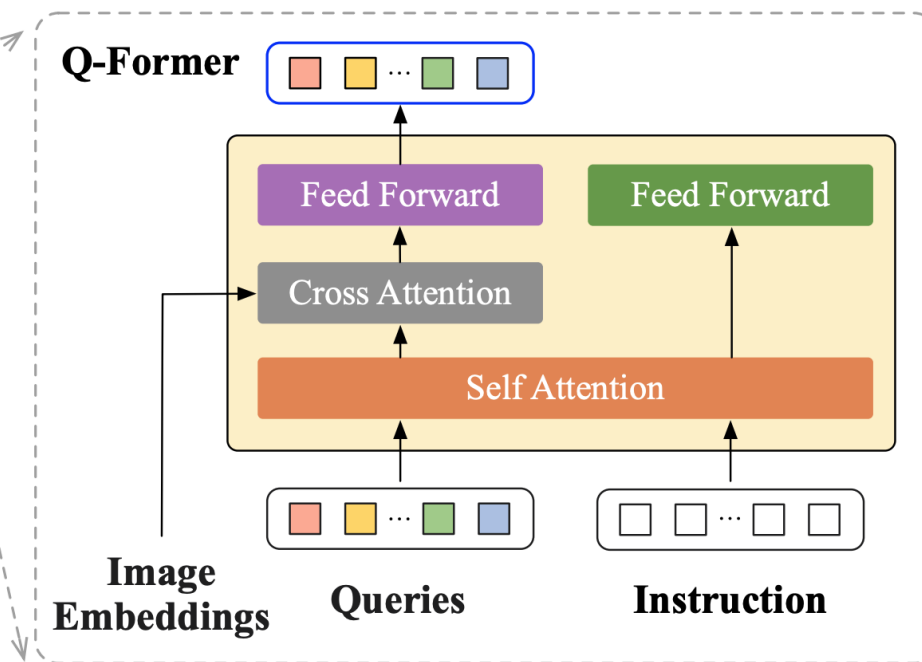
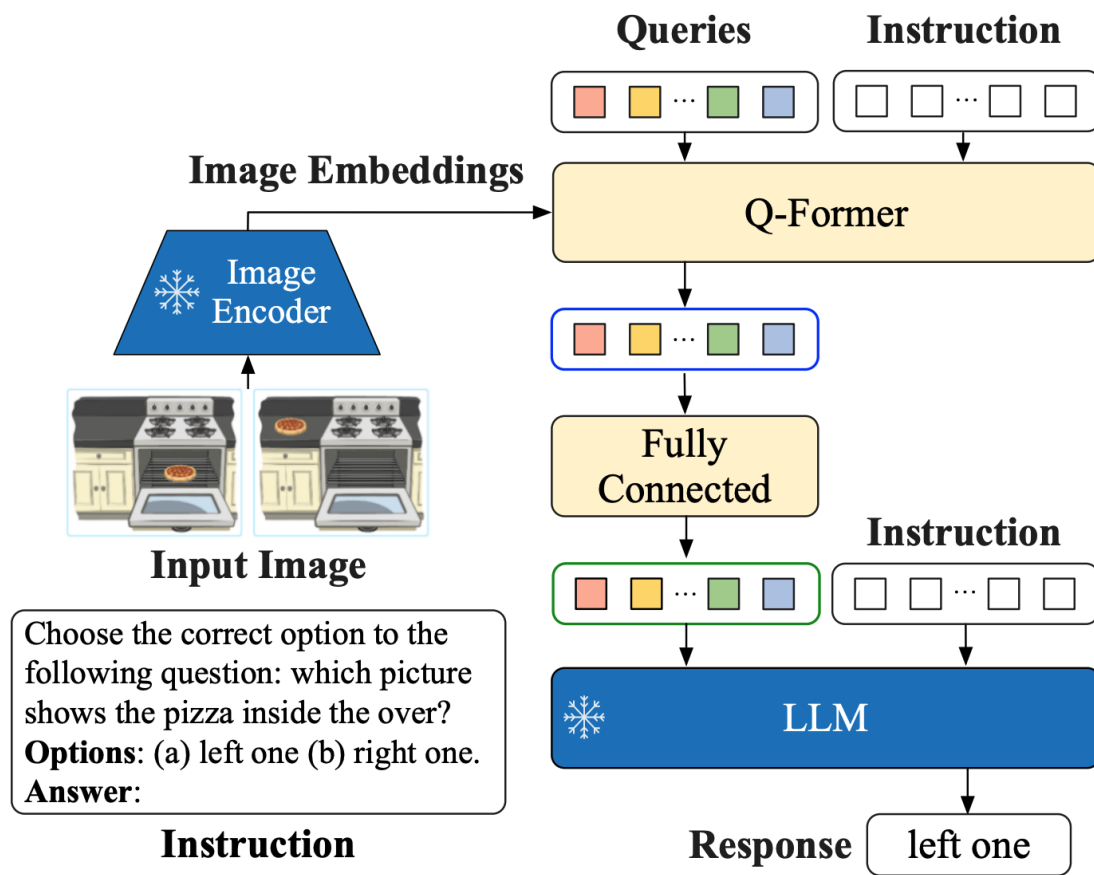
GPT-4: The unusual thing about this image is that a man is ironing clothes on an ironing board attached to the roof of a moving taxi.

BLIP and Flamingo supervision lead to descriptive responses instead of answering the instruction.

LLaVA

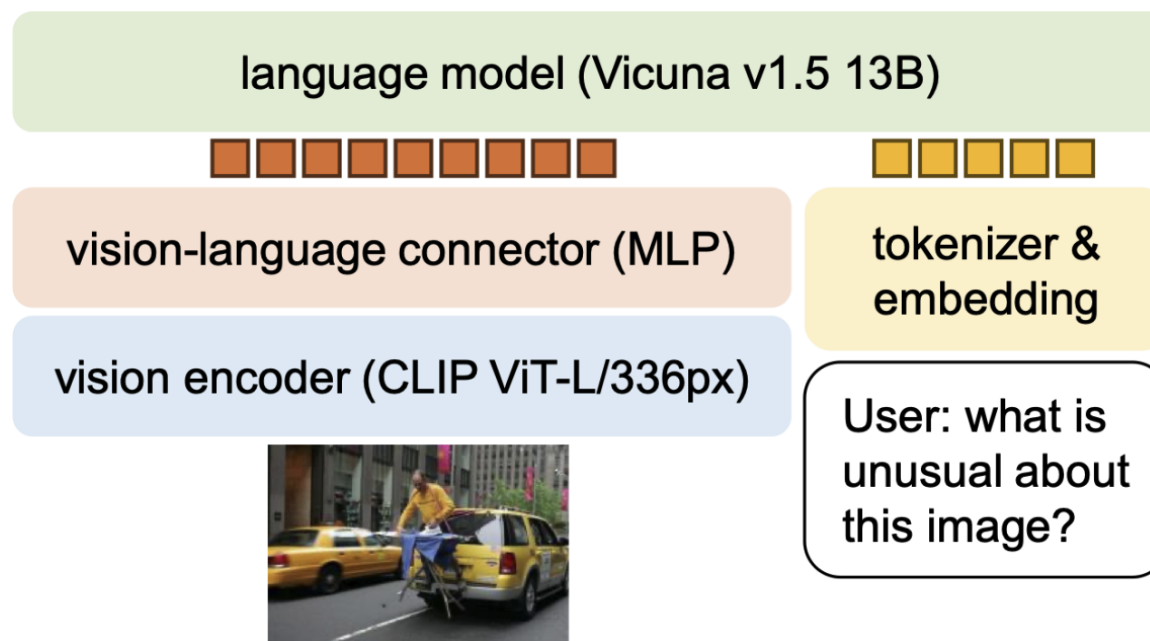
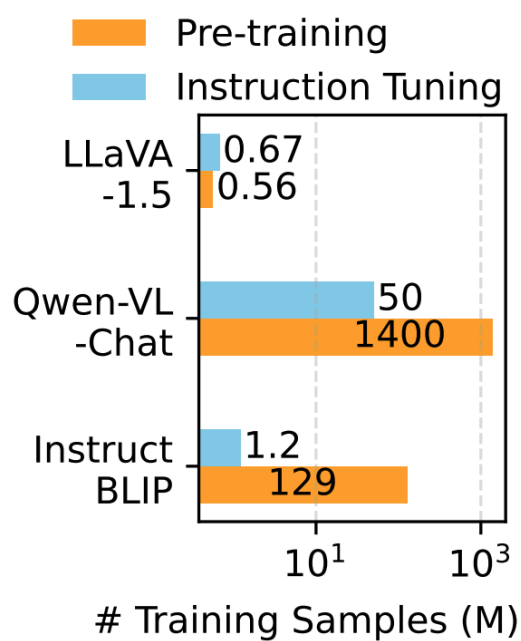
The unusual aspect of this image is a man ironing clothes on the back of a minivan or van. This is not a typical place to perform this activity, as one would usually iron clothes in a more stationary and safe location, such as a home, using a regular ironing board. The scene depicted in the image is peculiar as it involves a makeshift ironing setup on a vehicle, which can be both unsafe and unconventional. Additionally, it is not clear how the man is able to maintain balance and stability while ironing clothes in such an unstable environment.

InstructBLIP

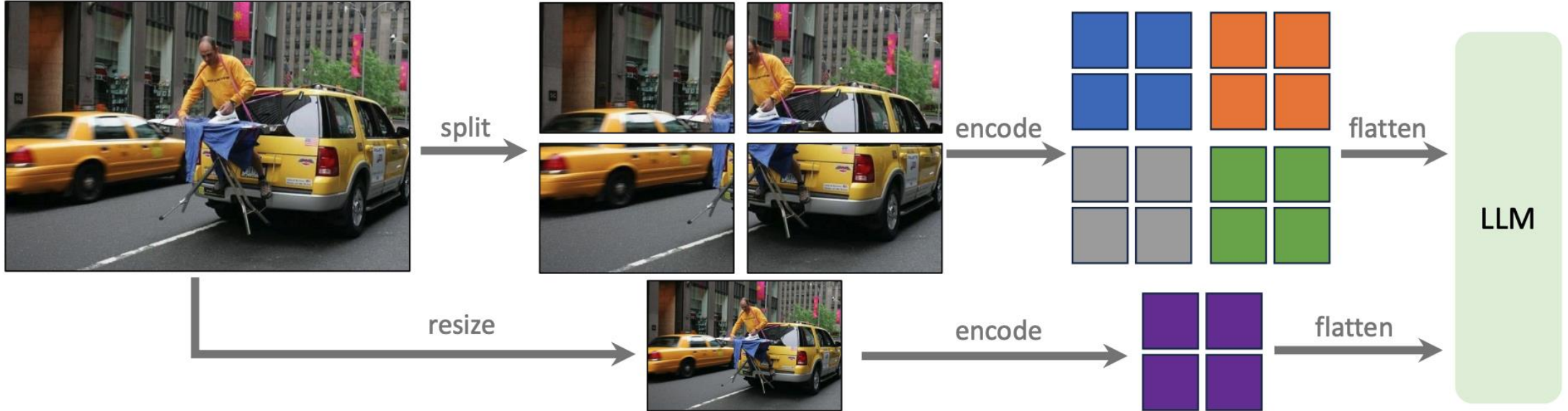


Instruction-aware Q-former module

LLaVA 1.5



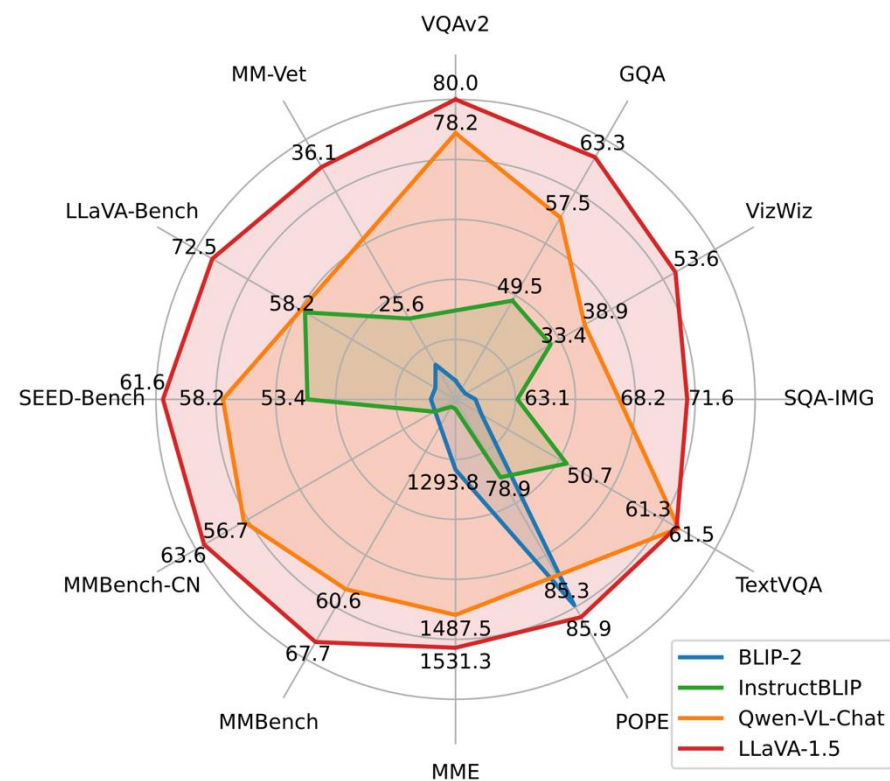
Scaling high resolution



Results

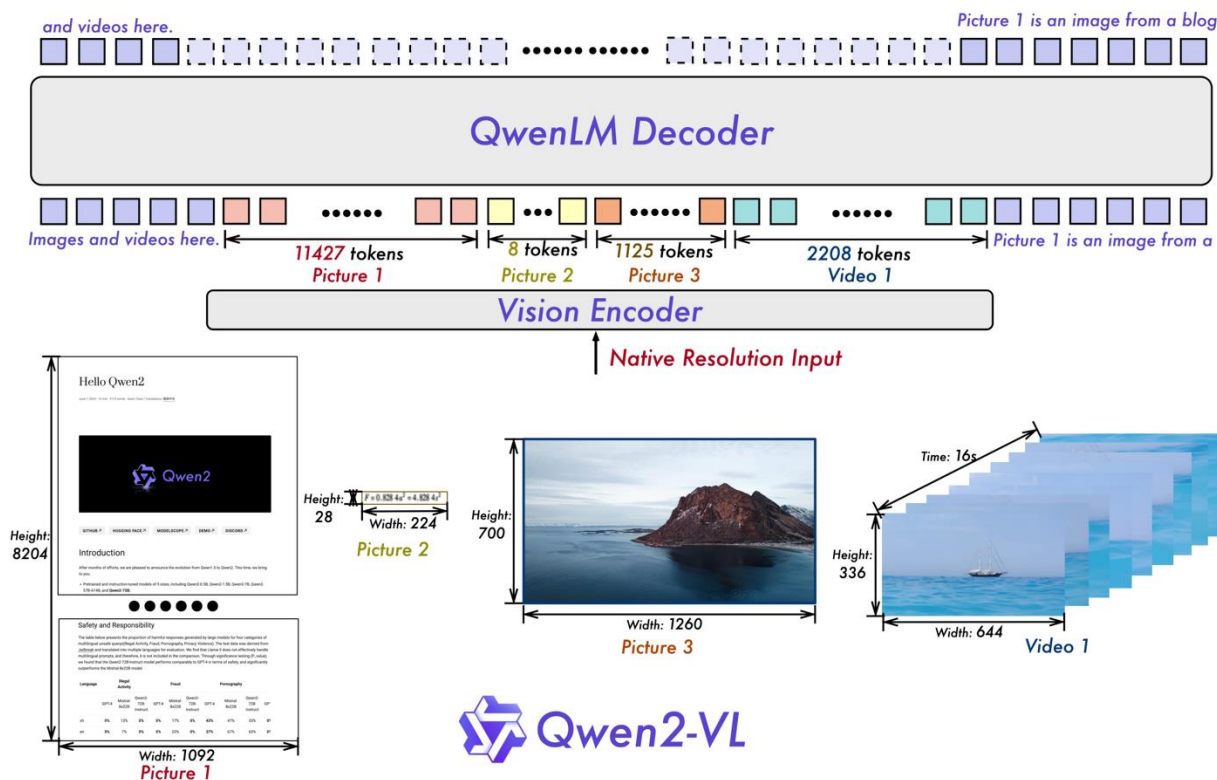
Method	LLM	Res.	GQA	MME	MM-Vet
InstructBLIP	14B	224	49.5	1212.8	25.6
<i>Only using a subset of InstructBLIP training data</i>					
0 LLaVA	7B	224	—	809.6	25.5
1 +VQA-v2	7B	224	47.0	1197.0	27.7
2 +Format prompt	7B	224	46.8	1323.8	26.3
3 +MLP VL connector	7B	224	47.3	1355.2	27.8
4 +OKVQA/OCR	7B	224	50.0	1377.6	29.6
<i>Additional scaling</i>					
5 +Region-level VQA	7B	224	50.3	1426.5	30.8
6 +Scale up resolution	7B	336	51.4	1450	30.3
7 +GQA	7B	336	62.0*	1469.2	30.7
8 +ShareGPT	7B	336	62.0*	1510.7	31.1
9 +Scale up LLM	13B	336	63.3*	1531.3	36.1

Table 2. **Scaling results** on data, model, and resolution.

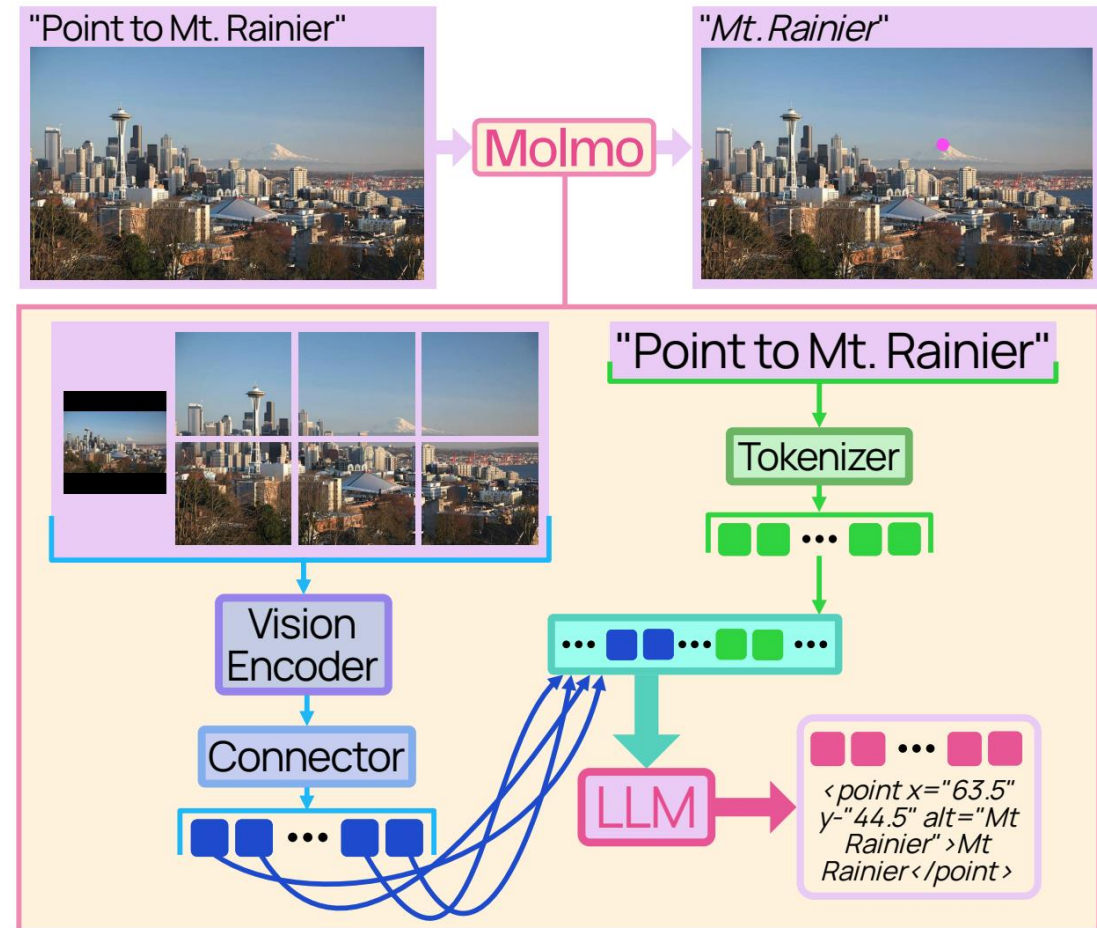
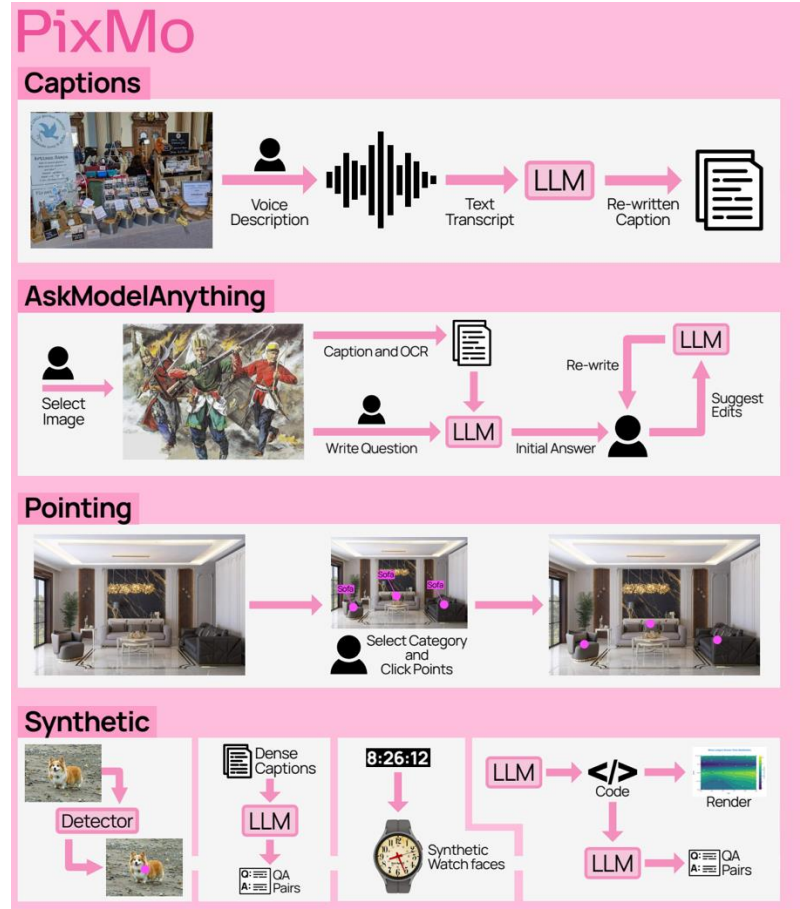


Qwen2-VL

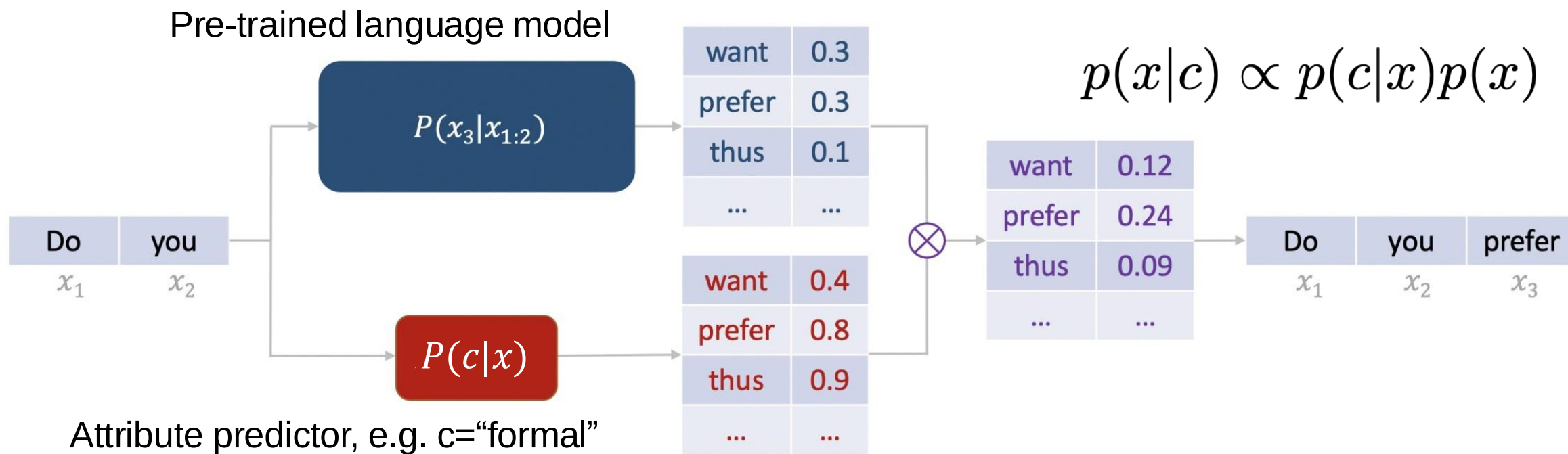
- **Dynamic Resolution:** Unlike before, Qwen2-VL can handle arbitrary image resolutions, mapping them into a dynamic number of visual tokens



Data is all you need?



No training: Conditioning via Bayes Rule

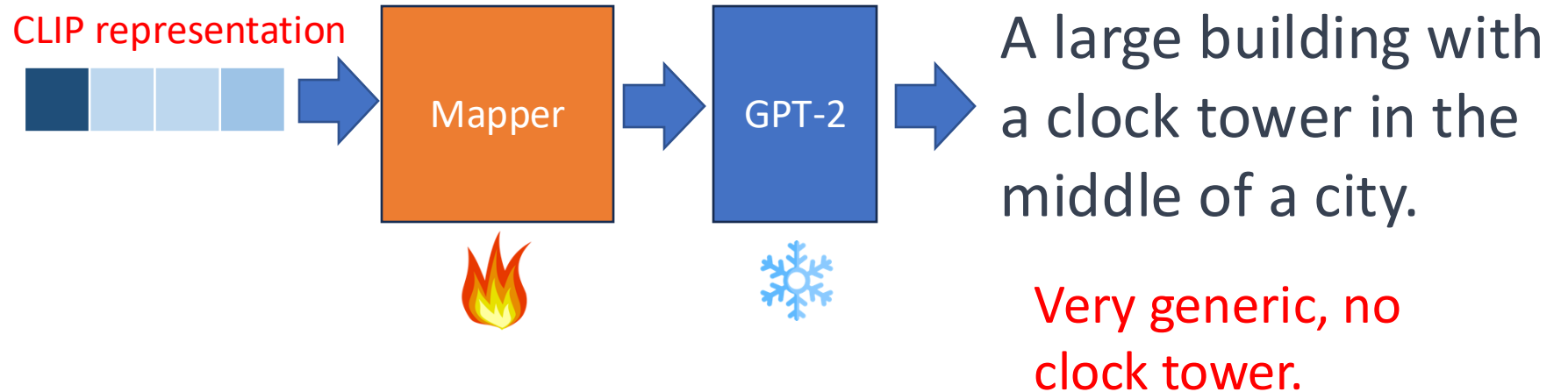
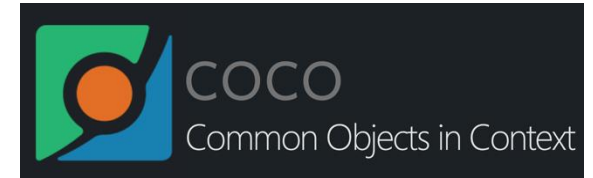


[Yang and Klein, FUDGE: Controlled Text Generation With Future Discriminators. NAACL 2021]

[Lazaridou et al. Multi-agent Communication Meets Natural Language. ACL 2020]

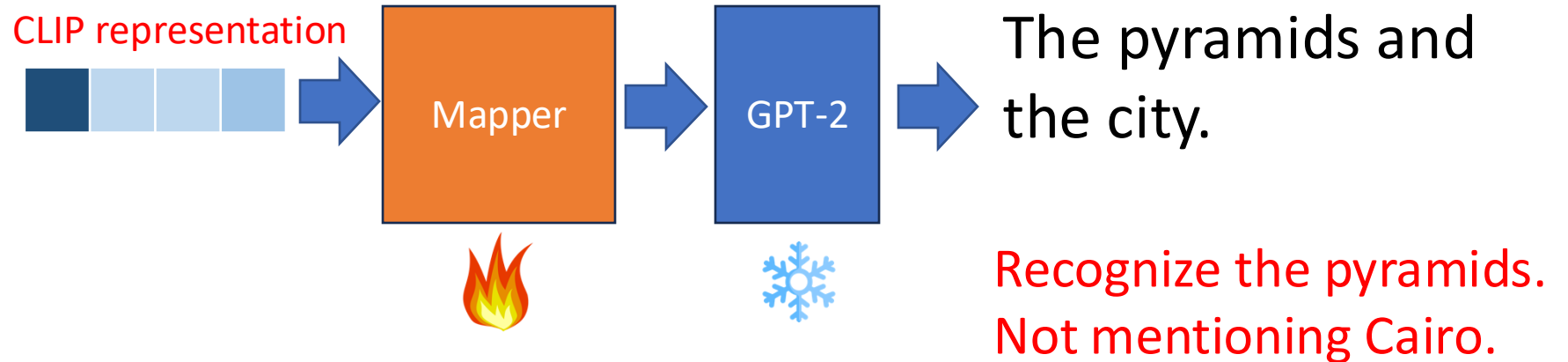
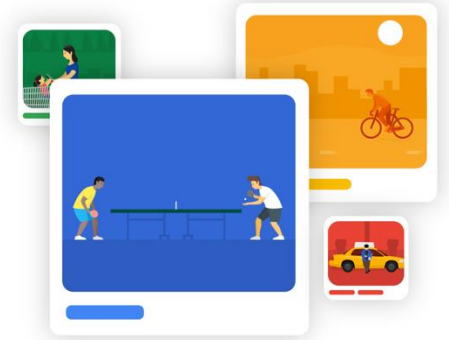
Fine-tune with COCO

- COCO contains **118K** images with associated captions
- Fine-tuning, once again, harms comprehension



Fine-tune with Closed Captions (CC)

- CC contains **3.3M** images with associated captions
- Adding data helps; but still limited vocabulary



We can use CLIP to re-rank the word logits

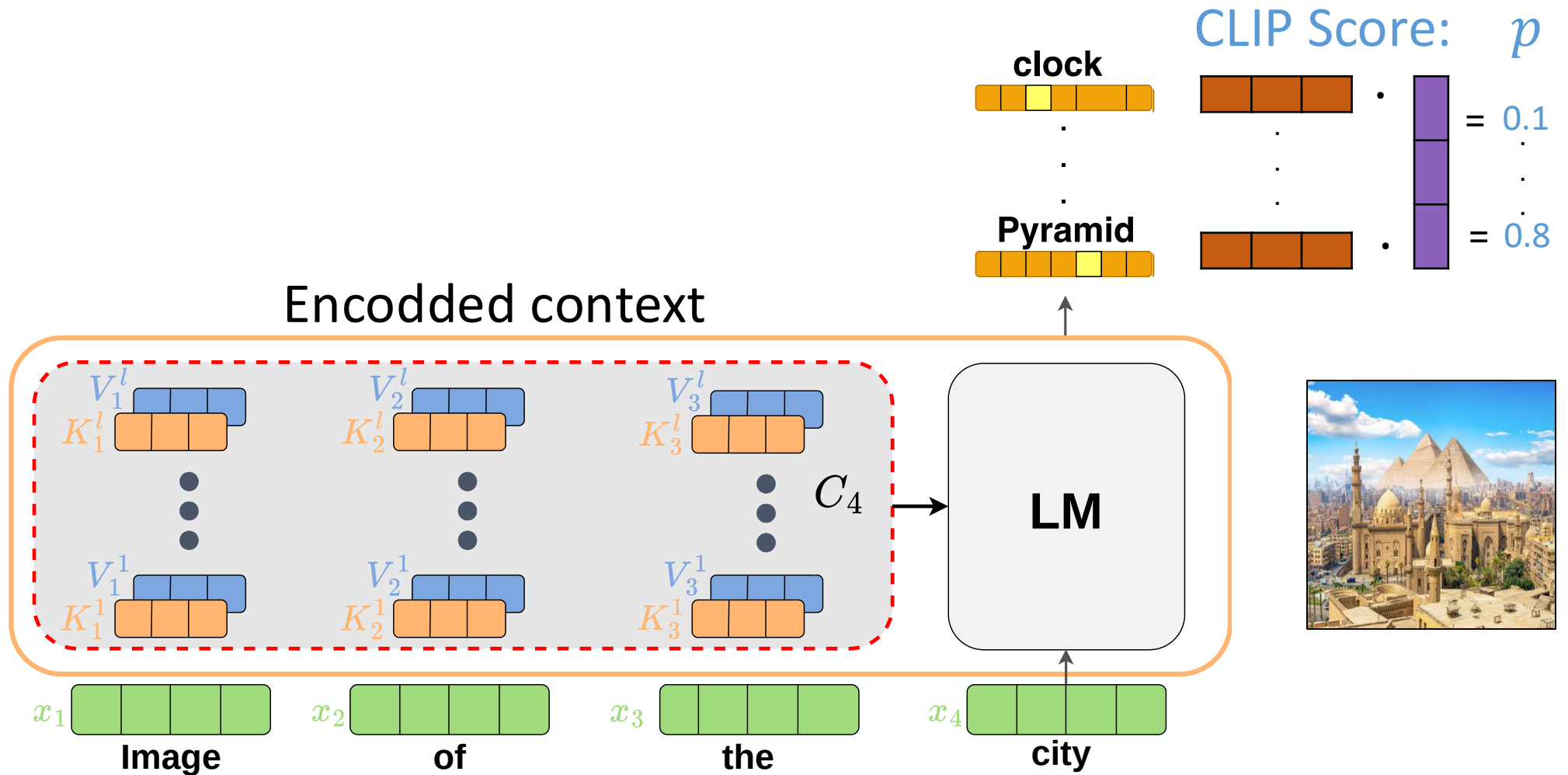
CLIP-Score( , "An Image of Cairo") = $\begin{bmatrix} \text{purple} \\ \text{purple} \\ \text{purple} \end{bmatrix} \cdot \begin{bmatrix} \text{brown} & \text{brown} & \text{brown} \end{bmatrix} = 0.9$

CLIP embeddings

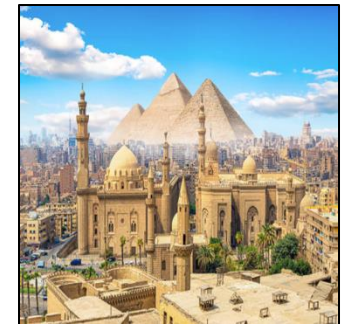
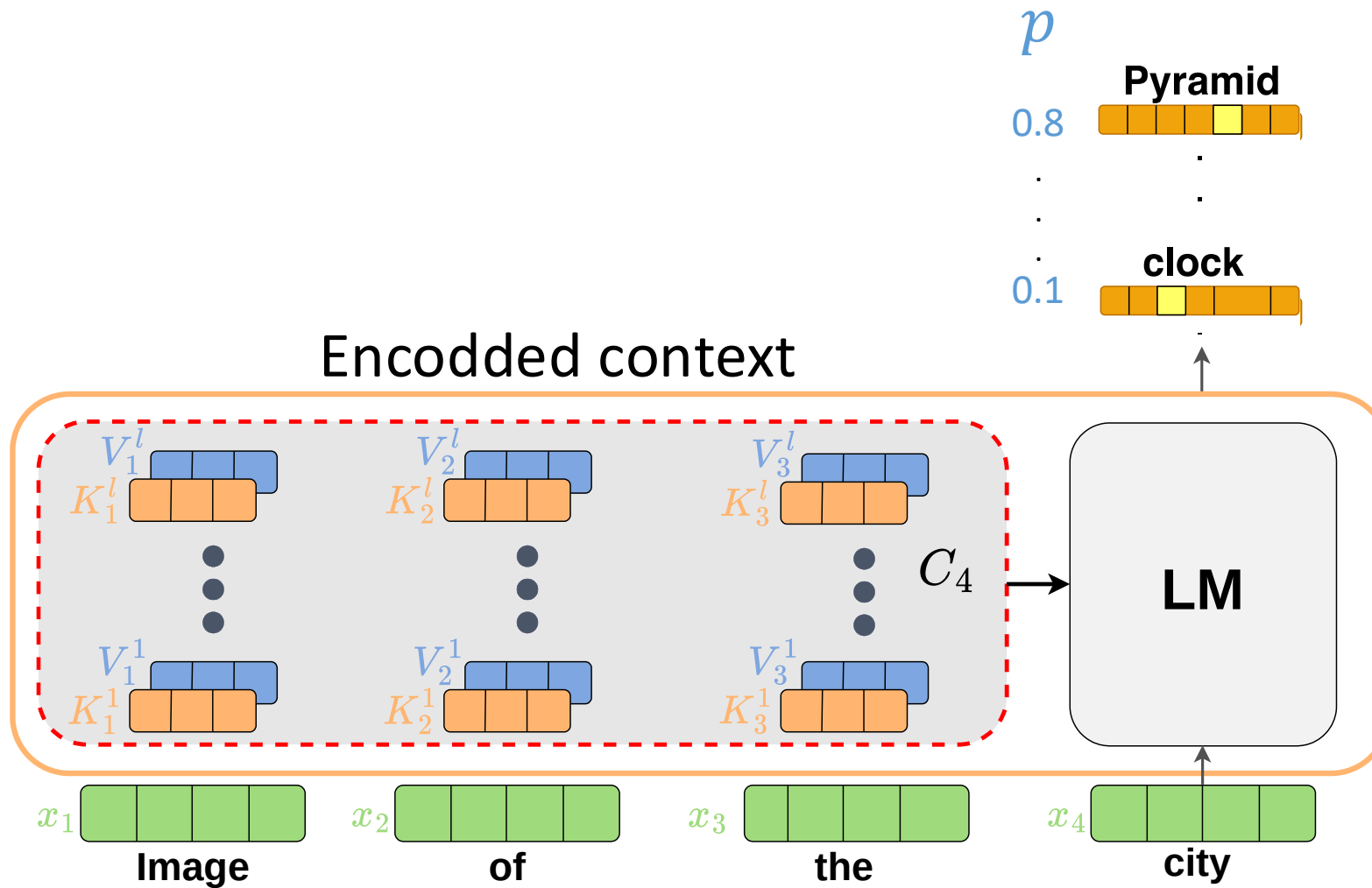
CLIP-Score( , "An Image of a Pyramid") = $\begin{bmatrix} \text{purple} \\ \text{purple} \\ \text{purple} \end{bmatrix} \cdot \begin{bmatrix} \text{brown} & \text{brown} & \text{brown} \end{bmatrix} = 0.8$

CLIP-Score( , "An Image of a clock") = $\begin{bmatrix} \text{purple} \\ \text{purple} \\ \text{purple} \end{bmatrix} \cdot \begin{bmatrix} \text{brown} & \text{brown} & \text{brown} \end{bmatrix} = 0.1$

Calculate clip score for top tokens



Re-rank the score



Issues:

1. **Cairo** is missing
2. To bring up Cairo we'll need to calculate CLIP score for more tokens which is computationally expensive
3. But also, relying only on CLIP for re-ranking will corrupt the language fluency

We'll deep dive next lecture



Pyramid



.

clock



Questions